Human Machine Informatics
Lecture 1:

Introduction

Kohei Nakajima

2026/10/07 Wed 10:25 - 12:10

Room 202

Course Description

How can we construct **autonomous systems** with human-like intelligence? This course approaches this question from the perspective of **nonlinear dynamics arising from the coupling of brain, body, and environment**. Building on insights from **embodied cognitive science**, the course explores a **constructive approach to intelligent systems** through an introduction to cutting-edge concepts and technologies in **complex systems, chaotic dynamical systems, physical reservoir computing, soft robotics, and neuromorphic devices**.

- Embodiment
- Where does spontaneity(=ghost) come from?

Where I'm from…

(education)

Ph. D. The University of Tokyo, 2009.
M. S. The University of Tokyo, 2006.
B. S. The University of Tokyo, 2004.

(major)
nonlinear dynamics,
chaos theory,
recurrent neural network,
embodiment.



Takashi Ikegami

(research experience)

2009 **Postdoctoral Researcher** (EU project: OCTOPUS)
Department of Informatics,
University of Zürich

2013 **JSPS Postdoctoral Fellow**
ETH Zürich

2014 **Assistant Professor**
The Hakubi Center for Advanced Research,
Kyoto University
(from 2015, 10, JST PRESTO Researcher)

2017 **Project Associate Professor**
The University of Tokyo

2020 **Associate Professor**
The University of Tokyo



Rolf Pfeifer

(topic)
soft robotics, morphological computation,
(physical) reservoir computing.

Octopus arm computer

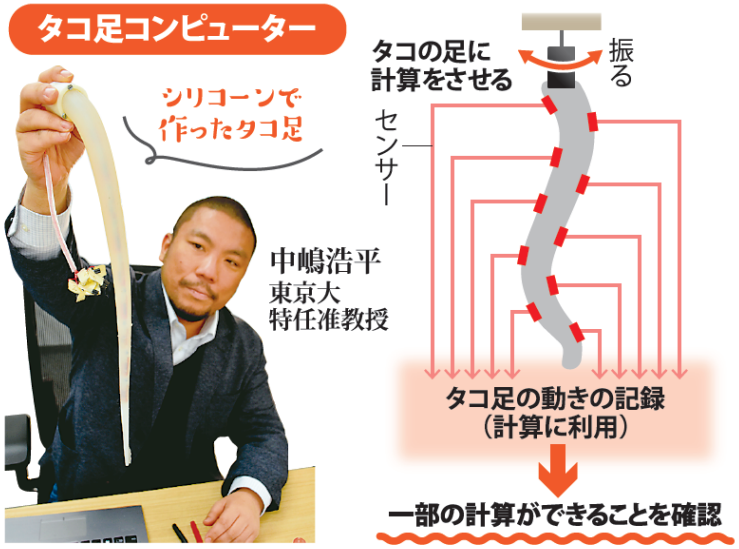
日経サイエンス「人工知能」（2020年6月）



Swissinfo 2013



毎日新聞2020





タコとイカ 異質な知性と体に秘められた謎|
ガリレオX第228回 (2020年9月27日)

https://www.youtube.com/watch?v=rn8wKflrm7Y

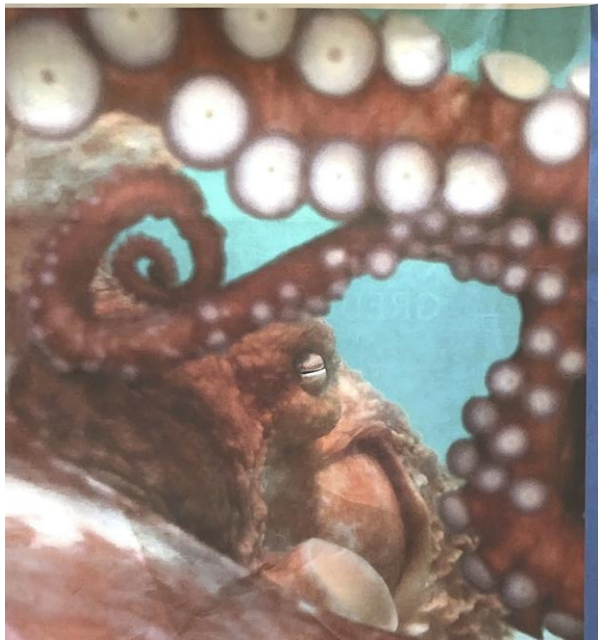
こんな日曜日が待ち遠しい。
NIKKEI The STYLE

日経新聞
2023/06/04





Mysterious Octopus



海の異才
未知なるオクトパス

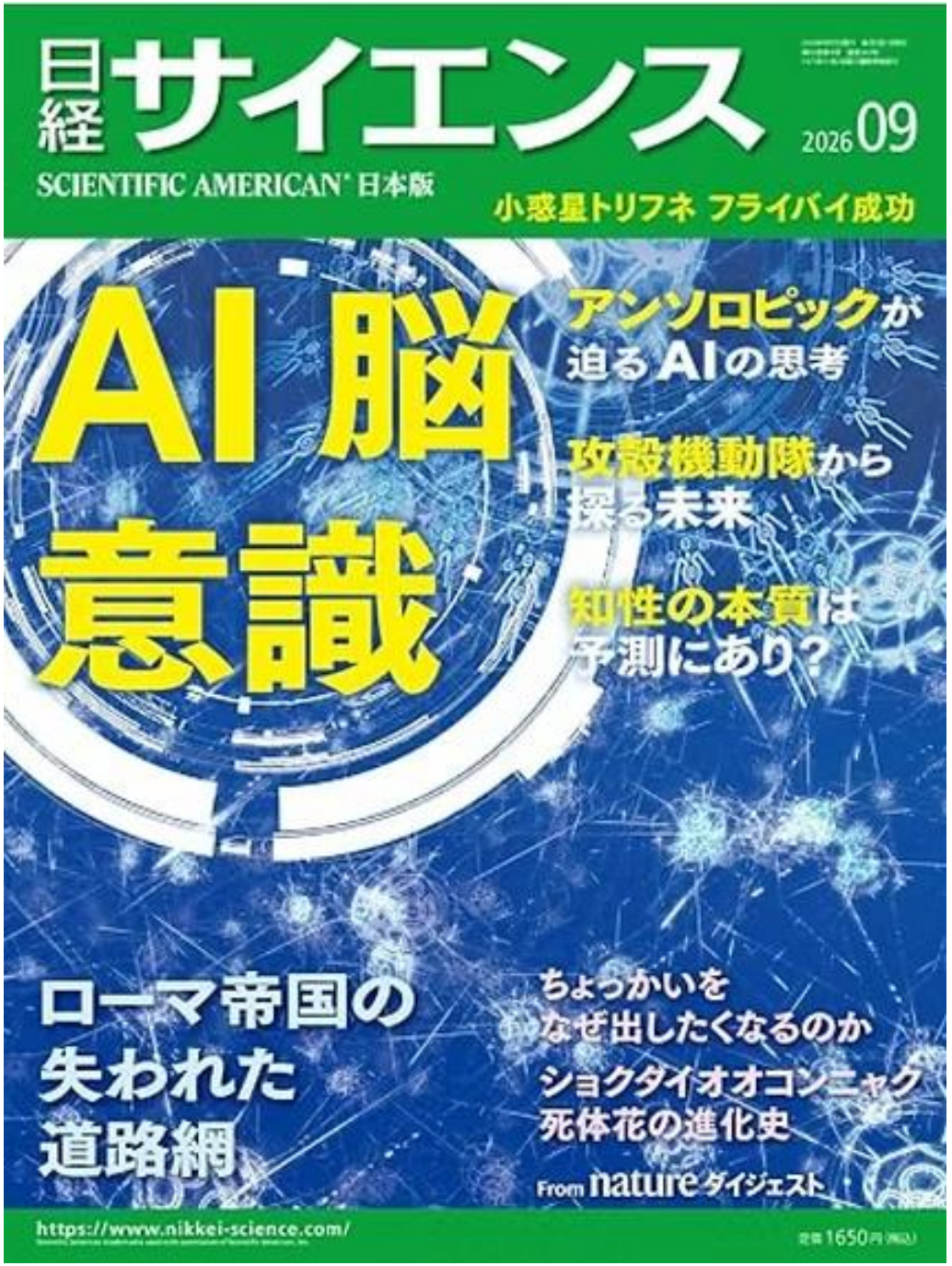
ある時は食材として食卓を彩り、
ある時はゆるキャラとなって見る者の心を癒やす。
遡れば、信仰や畏怖の対象でもあった。
人間と古いつきあいのあるタコは、
9つの脳と3つの心臓を持ち、8本の手足を
自在に操るユニークな生き物だ。
高い知性と、その秘められた能力に
関心が集まっている。

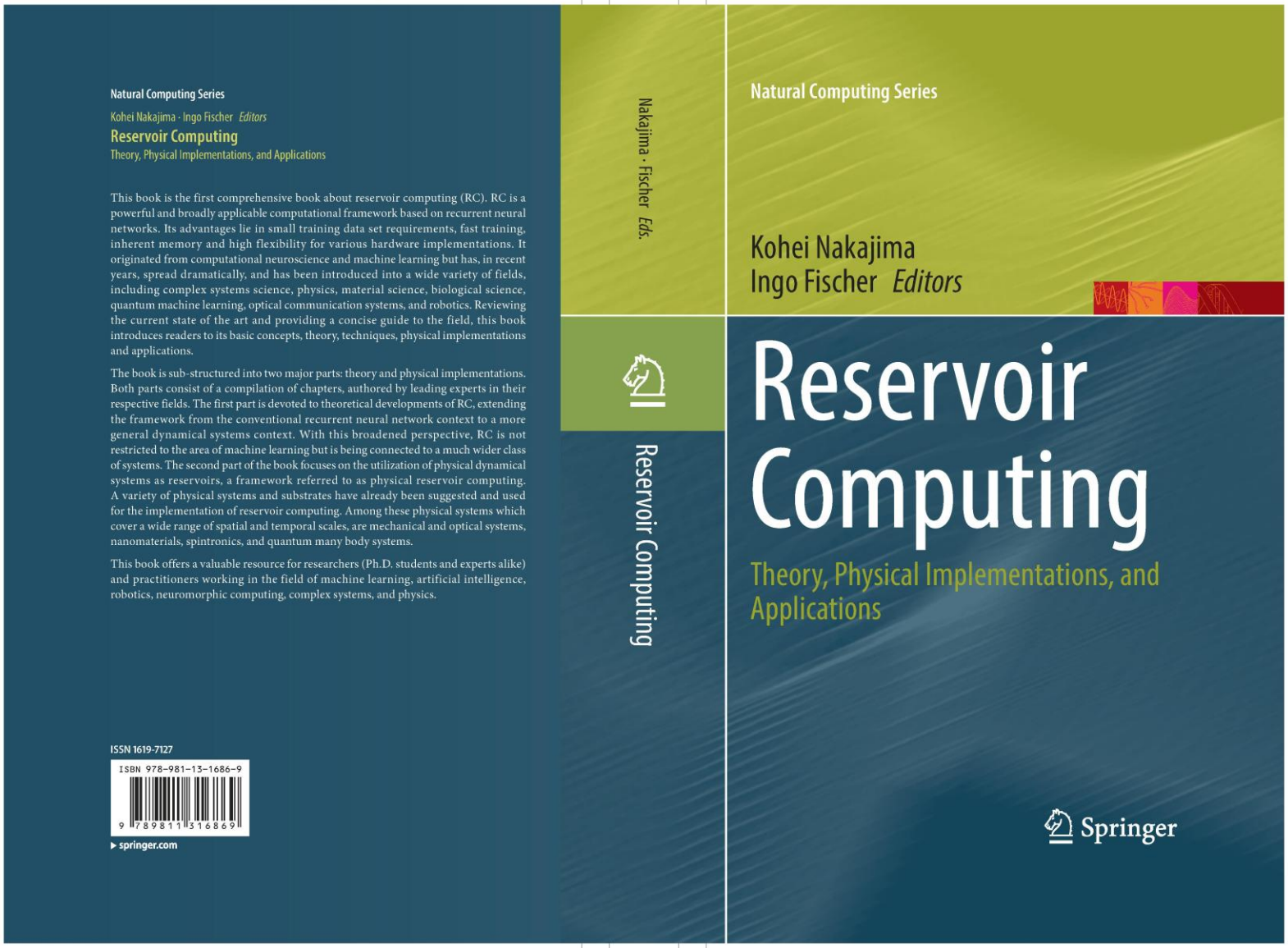
ミズダコは世界最大のタコで、大きいものは足を広げた長さが3メートル、体重は30キロ以上になる(同)



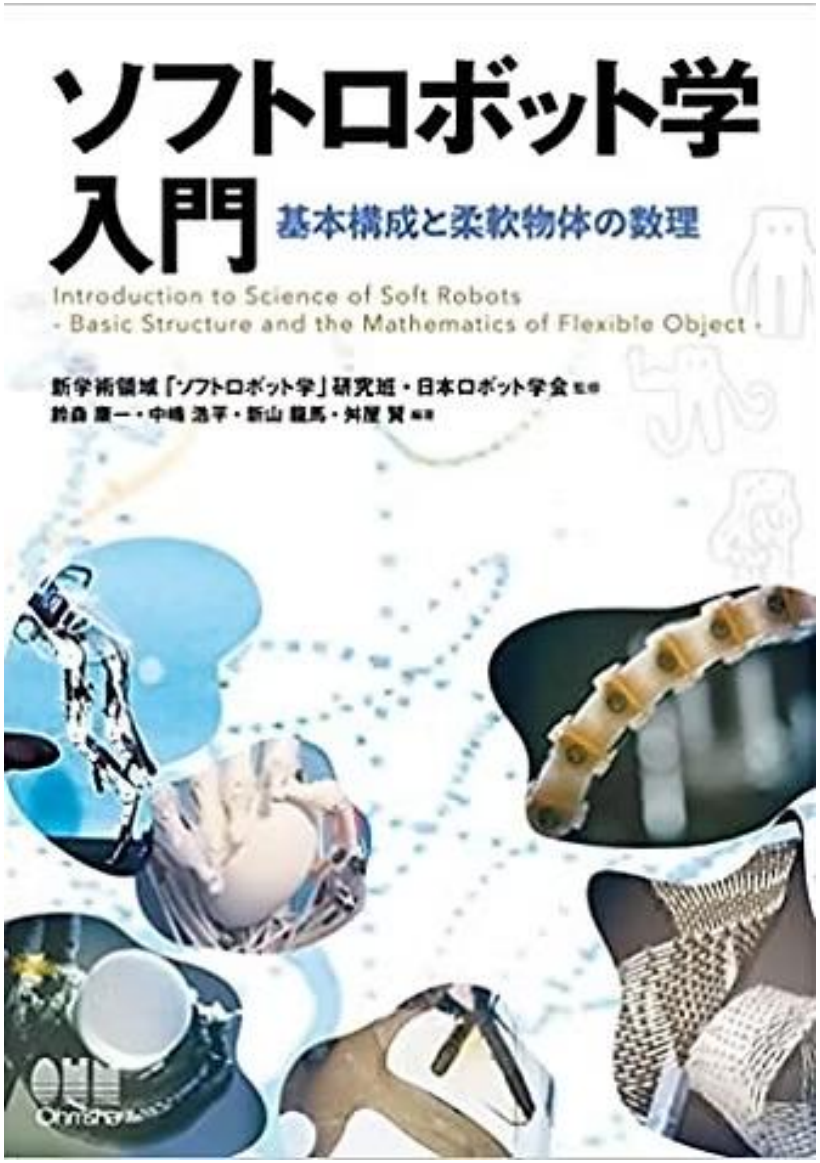
日経サイエンス2026年9月号「特集：AI・脳・意識」にて、大規模言語モデルにおける過渡カオスの研究が紹介されました

info: https://www.nikkei-science.com/page/magazine/202609.html

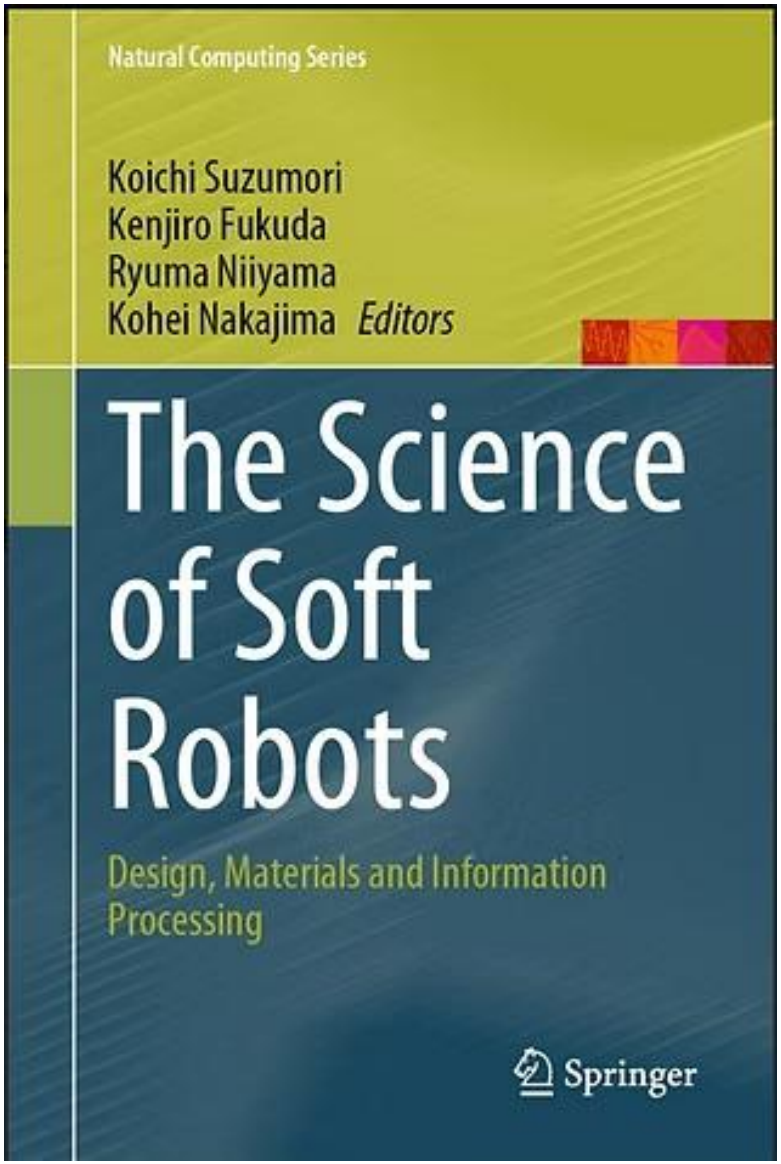




Nakajima, K., & Fischer, I. (Eds.). (2021). *Reservoir Computing: Theory, Physical Implementations, and Applications.* Springer Nature.



鈴森 康一、中嶋 浩平、新山 龍馬、舛屋 賢 編著、**ソフトロボット学入門 -基本構成と柔軟物体の数理-**、オーム社（2023年1月19日）. ISBN: 978-4-274-22998-5



Koichi Suzumori, Kenjiro Fukuda, Ryuma Niiyama, Kohei Nakajima, ***The Science of Soft Robots---Design, Materials and Information Processing***, Springer Singapore, 2023. ISBN: 978-981-19-5173-2

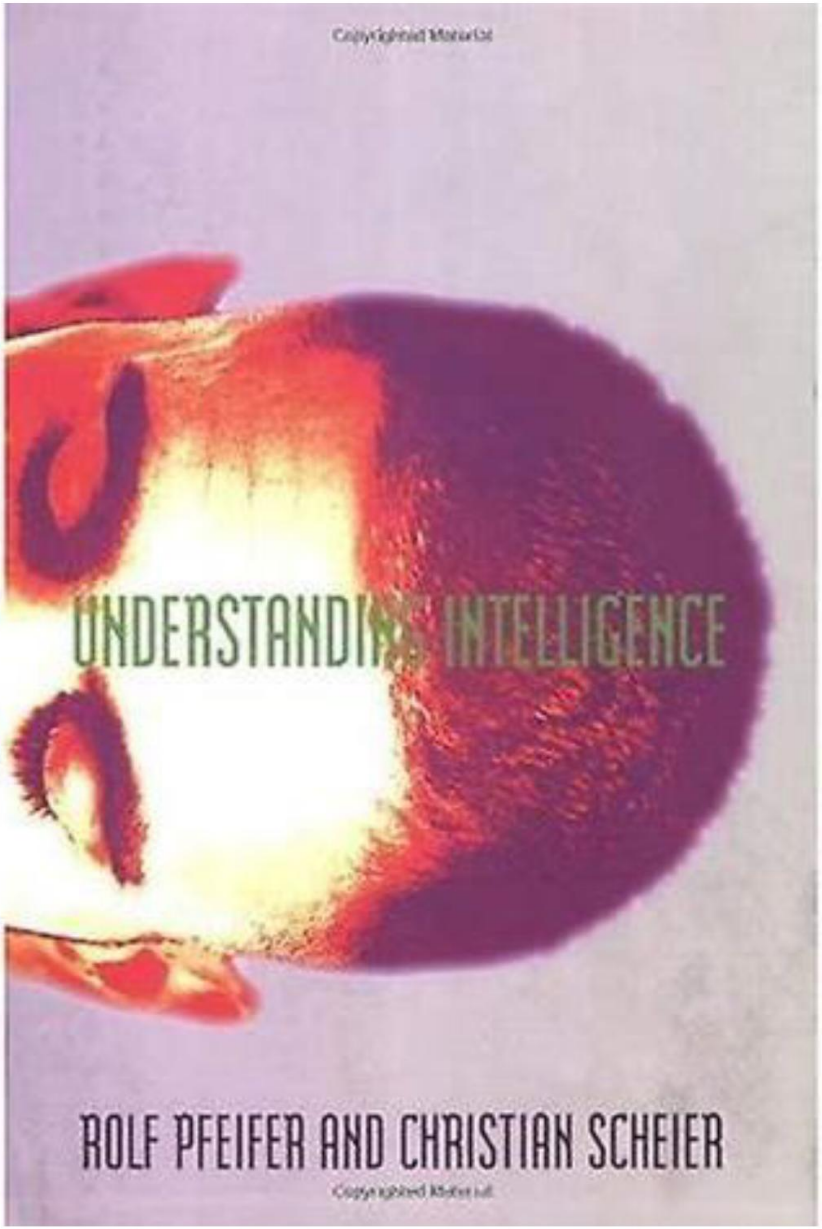
Schedule

1. Introduction
2. Dynamical system I: autonomous
3. Dynamical system II: input-driven dynamics
4. RC basics: echo state network
5. PRC basics: soft body dynamics and beyond
6. Memory and information processing capacity
7. Chaos in RC
8. Learning in RC
9. Self-organized criticality, Spontaneity, about Meaning

Schedule (tentative)

10. Dr. Katsuma Inoue (Univ. Tokyo):
“Dynamics that learns”
11. Dr. Tomoyuki Kubota (Univ. Tokyo):
“Reservoir computing generalized”
12. Dr. Nozomi Akashi (Kyoto Univ.):
“Embedding random dynamical systems”
13. Dr. Tran Hoc Quan (Fujitsu):
“Quantum reservoir computing”

Textbook: *Understanding Intelligence*



frame problem, symbol grounding problem, Braitenberg vehicles, embodiment, situatedness, autonomy, dynamical systems approach, sensor-motor coupling, recurrent neural networks, evolution of hardware, affordance, Randal Beer's active perception, Edelman's Darwin II, evolutionary robotics,…

Understanding Intelligent systems

- Embodiment: brain-body-environment

(it is not only for making robots…)

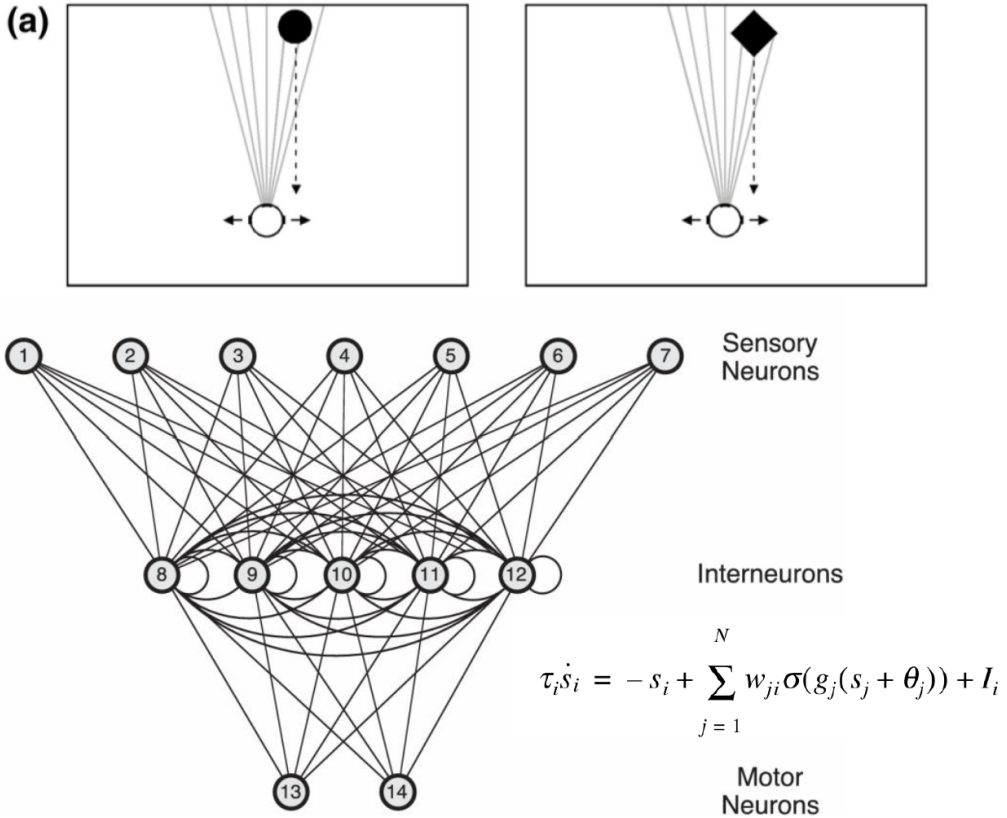
- Nonlinear dynamical systems

- Reservoir computing (RC) and physical reservoir computing (PRC)

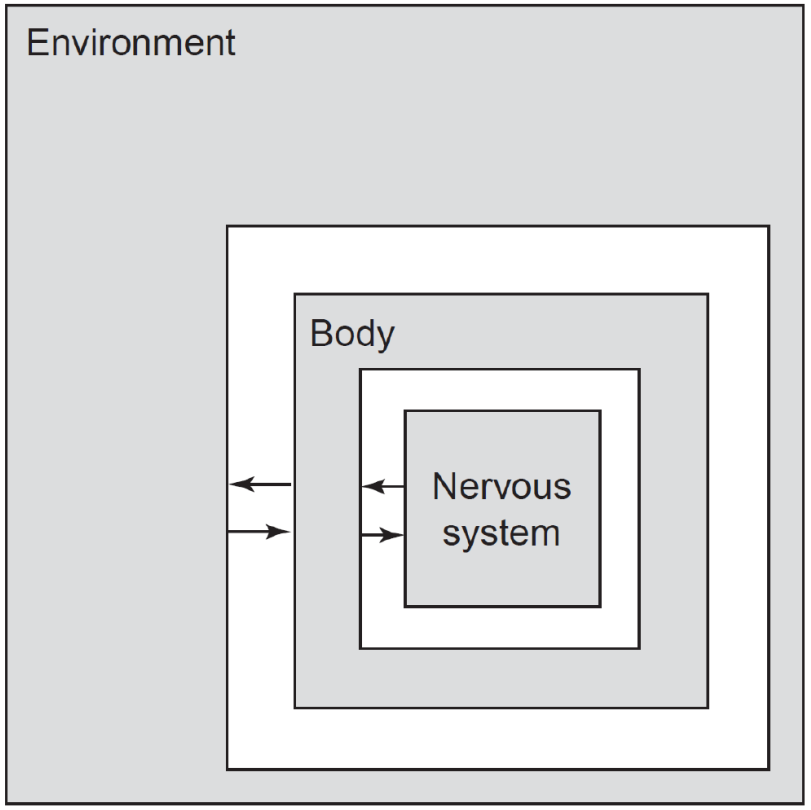
Bridging **physical dynamics** and its **information processing capability**!

Brain-body-environment system as dynamical systems (sensorimotor coupling)

Dynamical analysis: catching circles and avoiding diamonds



A dynamical perspective on a situated, embodied agent.

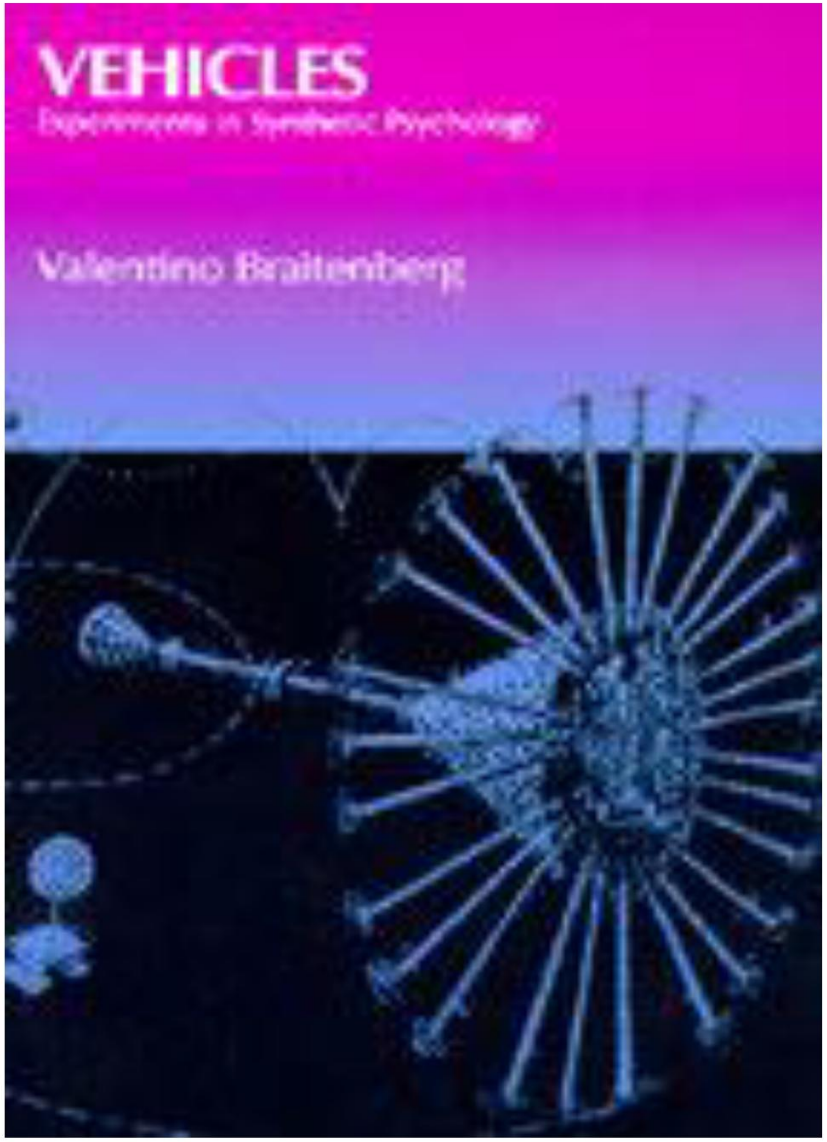


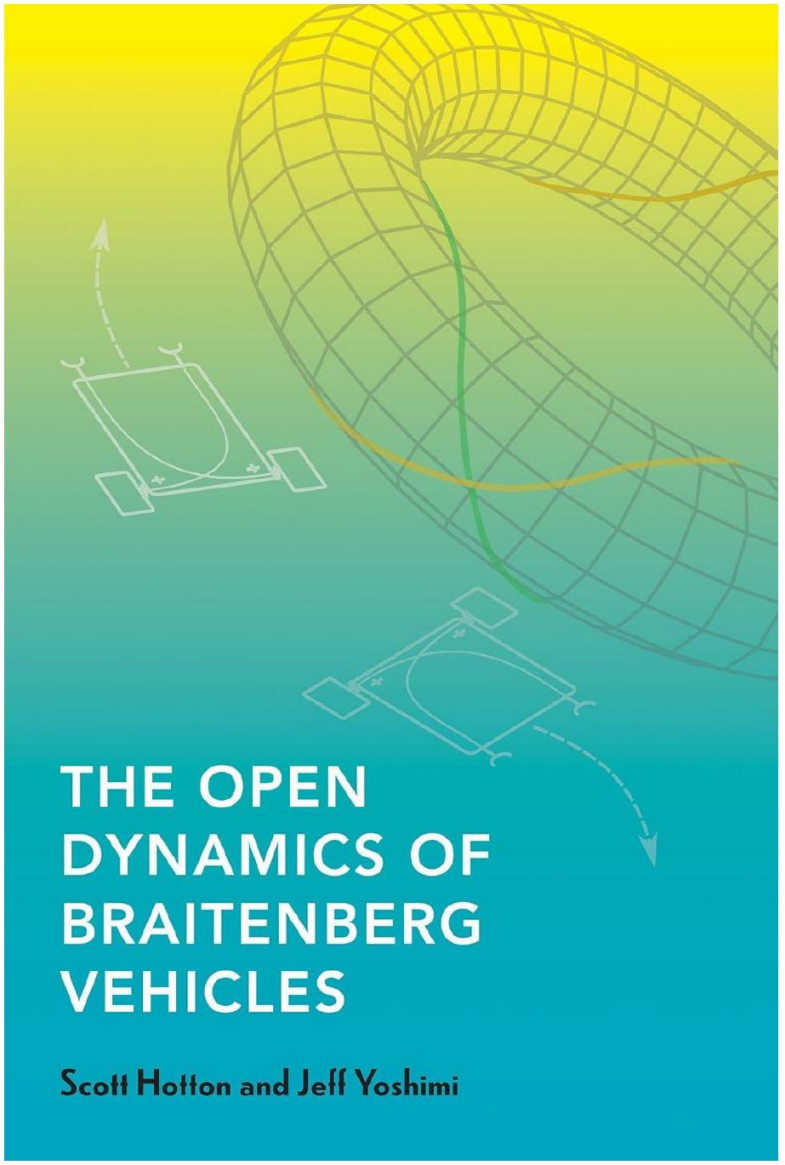
Beer, R. D. (2000). Dynamical approaches to cognitive science. Trends in cognitive sciences, 4(3), 91-99.

Evolution and Analysis of Brain-Body-Environment Systems, by Randall D. Beer (youtube.com)

Active categorization: categorization is not localized in the brain!

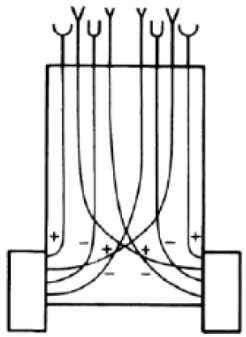
Braitenberg vehicles



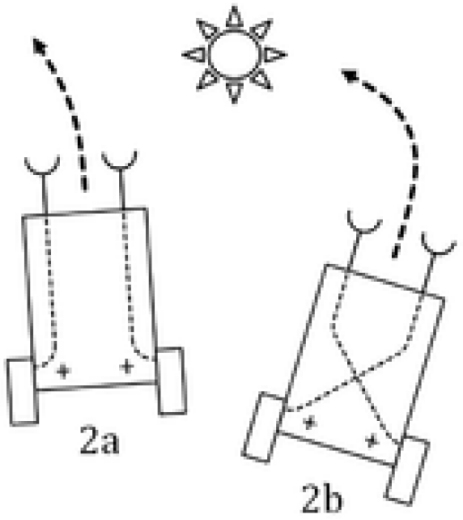


Braitenberg vehicles: examples

1. 回遊
2. 恐怖と攻撃
3. 愛情 reservoir
4. 価値と嗜好
5. 論理
6. 淘汰、見えざる手
7. 概念
8. 空間・物体・運動
9. 形状
10. 概念形成
11. 規則と規則性
12. 思考連鎖 chaos
13. 予見
14. 自我と楽観主義



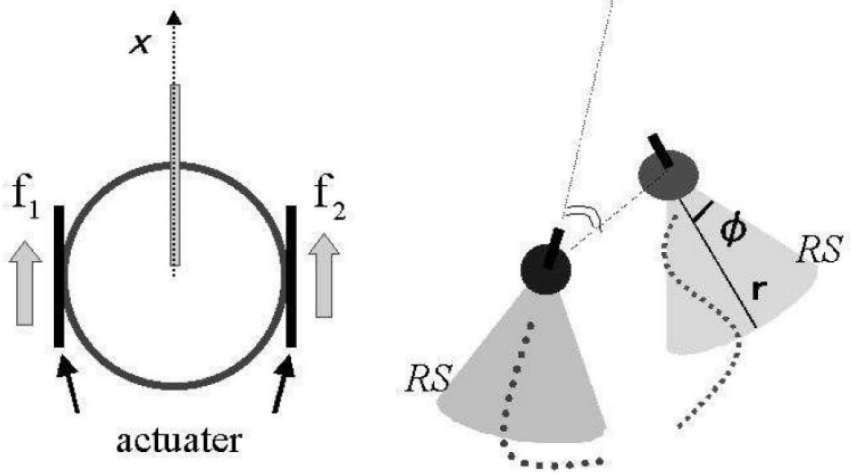
3c

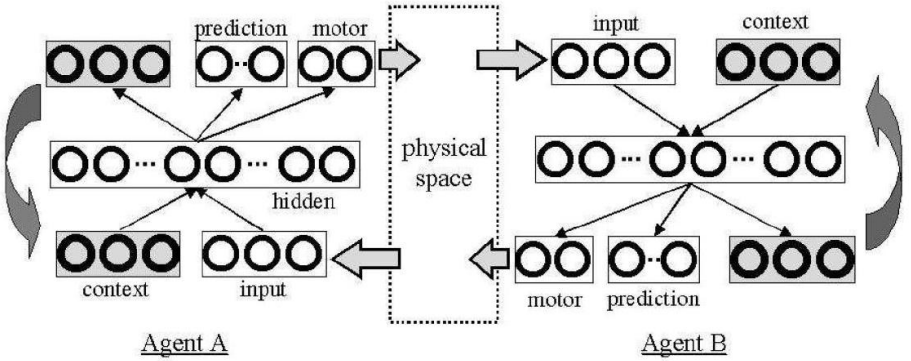


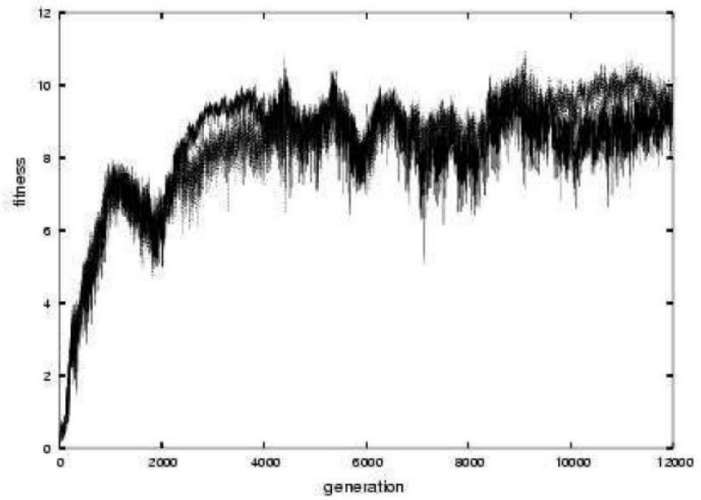
- In a complex environment, Braitenberg vehicles will exhibit diverse dynamic behavior through sensorimotor coupling.
- It even appear to be intelligent.

Vehicle 15: Turn taking 1

Hiroyuki Iizuka and Takashi Ikegami: Adaptability and Diversity in Simulated Turn-taking Behavior. Artificial Life 10: 361-378 (2004).







$$h_j(t) = g(\sum_i w_{ij} y_i(t) + \sum_l w'_{lj} c_l(t-1) + b_{j1}),$$

$$z_k(t) = g(\sum_j u_{jk} h_j(t) + b_{j2}),$$

$$c_l(t) = g(\sum_l u'_{jl} h_j(t) + b_{j3}),$$

$$g(x) = 1/(1+\exp(-x)),$$

- Using GA to tune network weights to maximize a turn taking frequency!

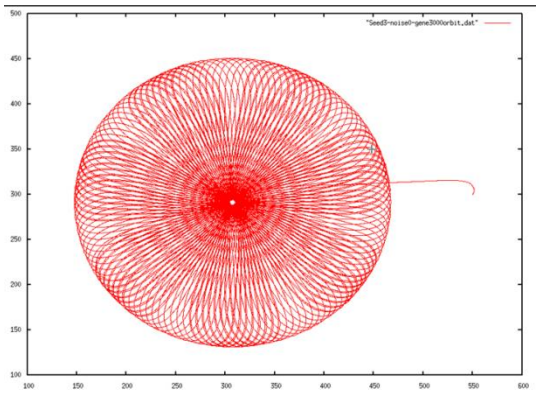
Vehicle NO.15: turn taking

Regular T.T.

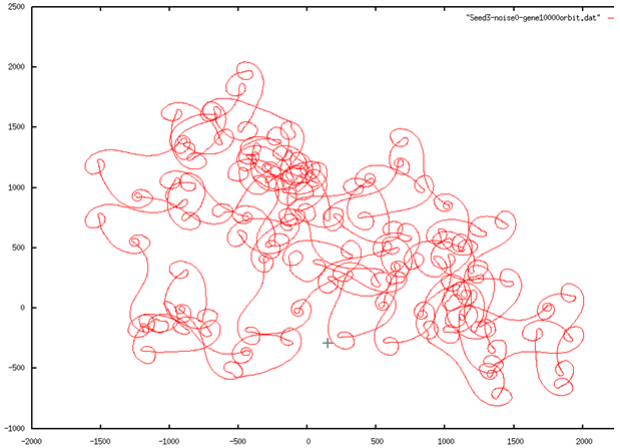
Chaotic T.T.







Trajectories



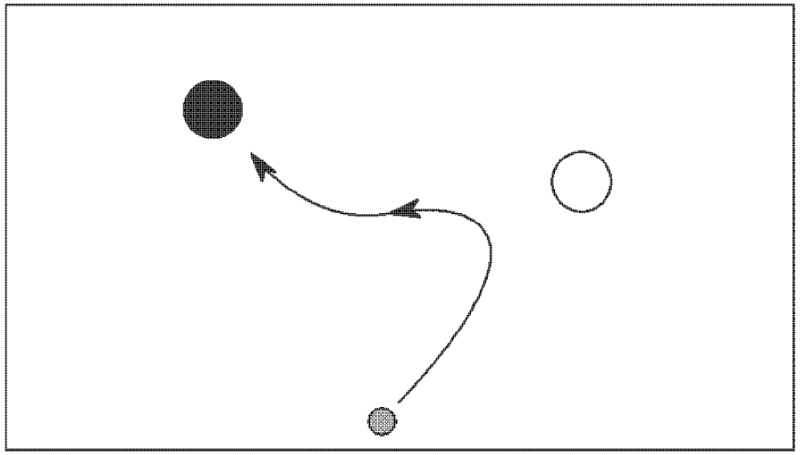
Rule-based interaction

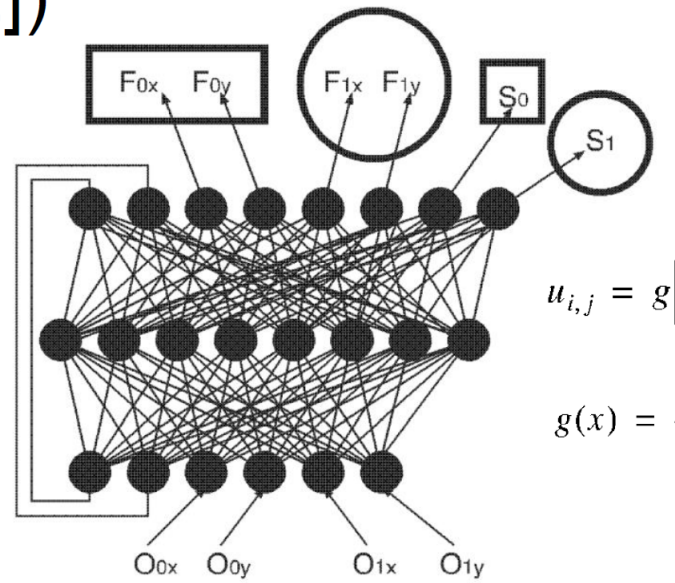
Dynamic interaction

Modeling microslip 1

Yuta Ogai and Takashi Ikegami: Microslip as a Simulated Artificial Mind. *Adaptive Behavior.* 16(2-3): 129-147 (2008)

- Microslip: fluctuation of behaviors (making instant coffee [Reed et al.,1992])





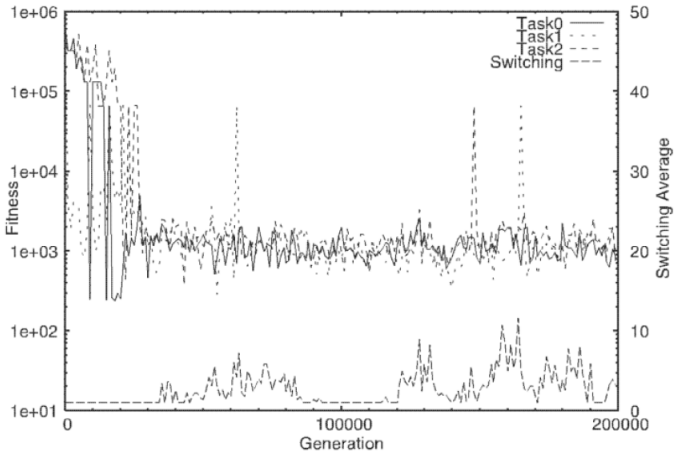
$$u_{i,j} = g\left(\sum_{k=1}^{N_{i-1}} w_{i-1,j,k} u_{i-1,k}\right).$$

$$g(x) = \frac{1}{1+e^{-\beta x}}.$$

Task 0: A single object 0 is in the field, and the agent has to obtain it.

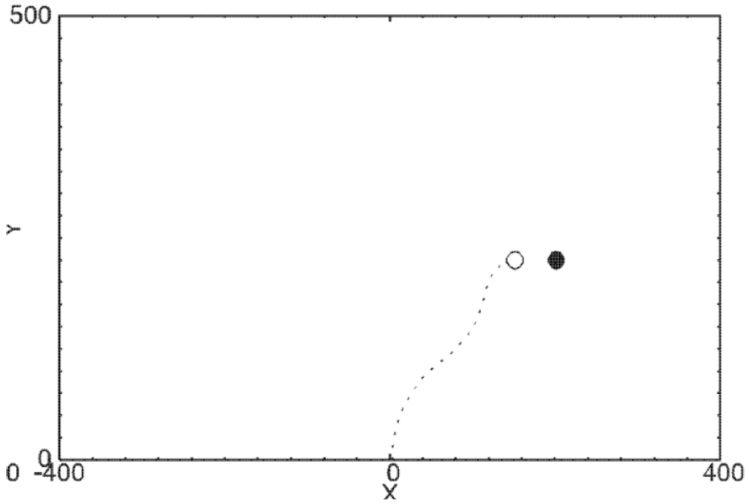
Task 1: A single object 1 is in the field, and the agent has to obtain it.

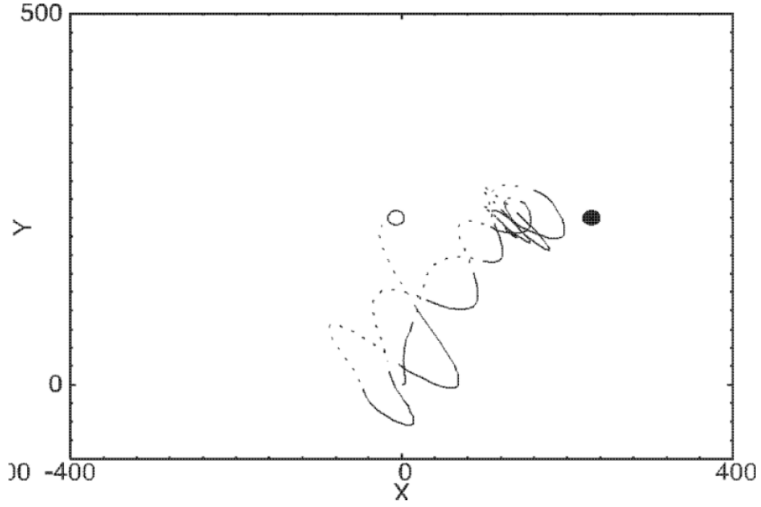
Task 2: Two objects are in the field, and the agent has to obtain one of the two.

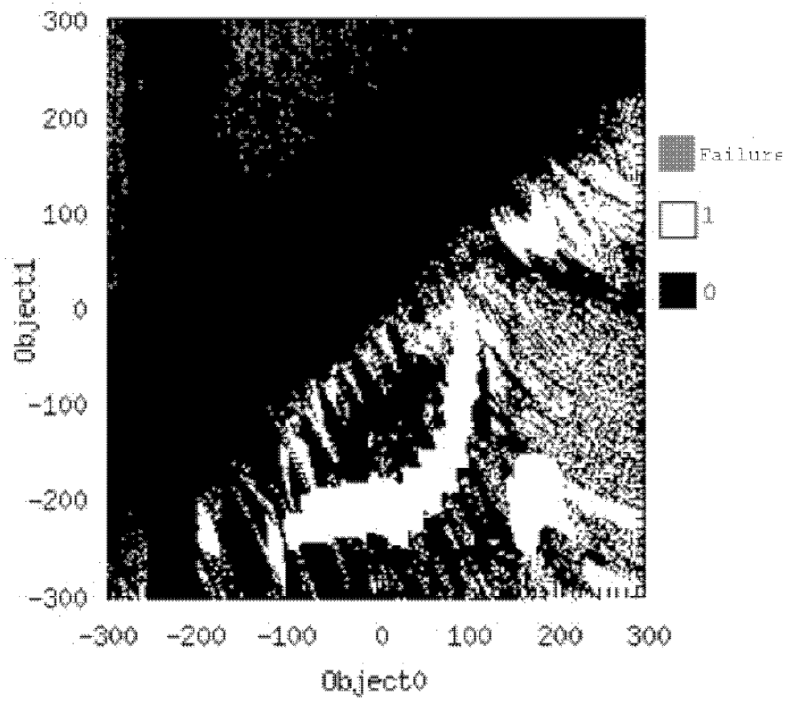


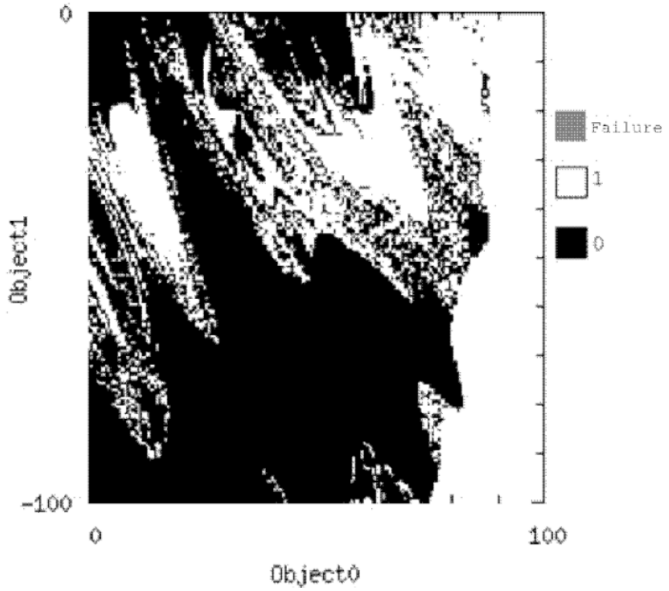
Modeling microslip 2

Yuta Ogai and Takashi Ikegami: Microslip as a Simulated Artificial Mind. *Adaptive Behavior.* 16(2-3): 129-147 (2008)



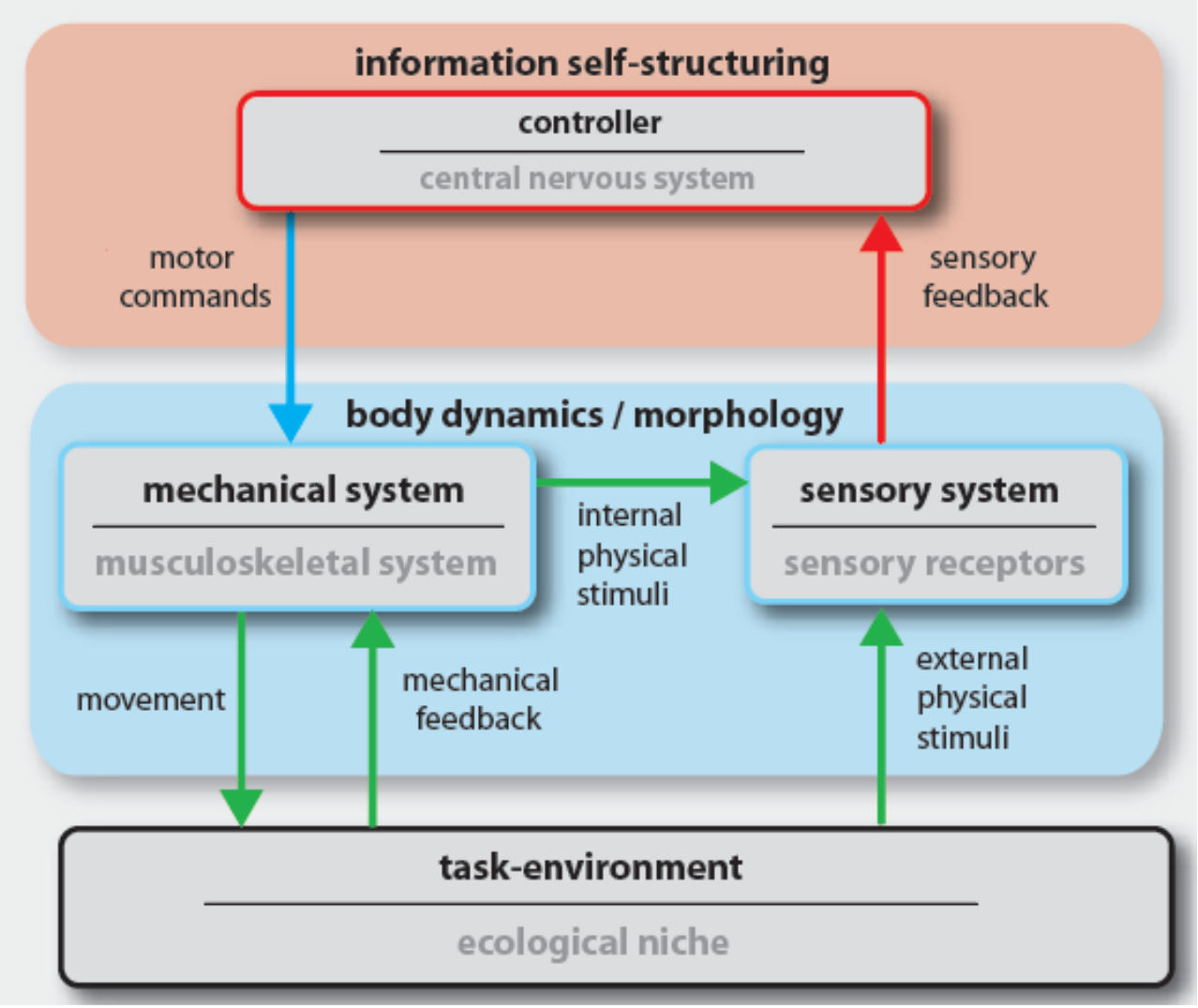






- Fractal basin appeared!

Embodiment



(R. Pfeifer et al., Science, 2007)

The behavior of robots (or animals) is generated by the dynamic coupling between the controller (brain), the body, and the environment!

Dead trout

2024 Ig Noble Prize winner!

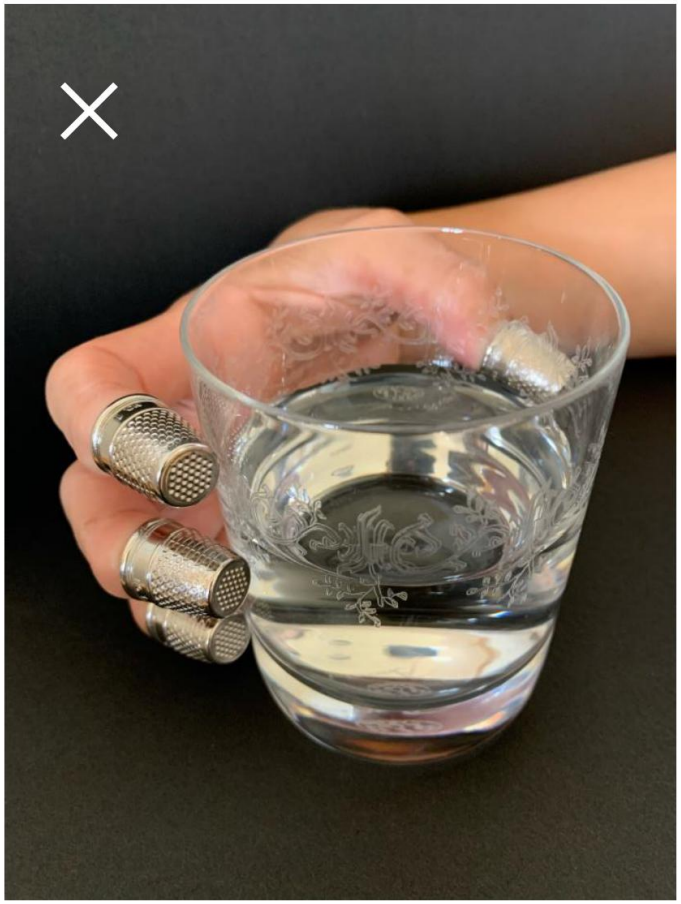


Liao, J. C. (2004). *Journal of Experimental Biology* **207**(20): 3495-3506.
Liao, J. C., et al. (2003). *Science*, 302(5650), 1566-1569.

- Exploiting body morphology, fluid dynamics, and resulting repulsive force induced by vortex, dead fish can “swim”!

Morphological computation





Suzumori, K., Fukuda, K., Niiyama, R., & Nakajima, K. (Eds.). (2023). The Science of Soft Robots: Design, Materials and Information Processing. Springer Nature.

- Collision detection is naturally solved in the physical world!
- Functionality is outsourced to the softness!

Soft robotics: Robots made of soft materials

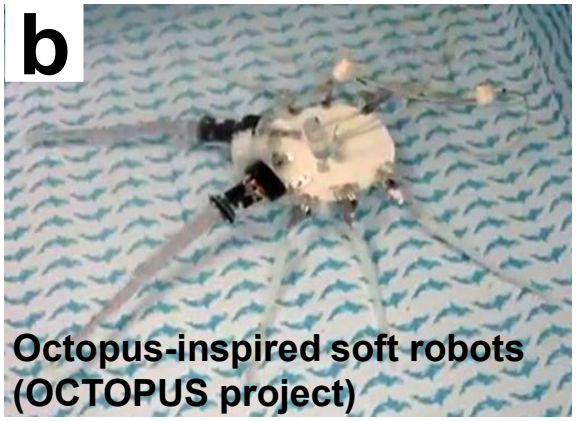
Octopus



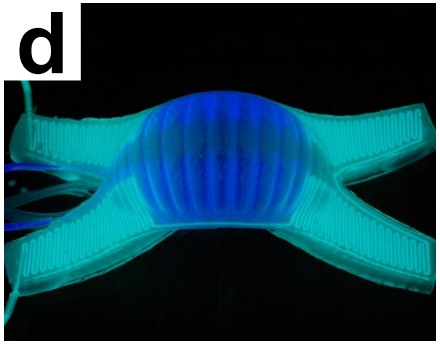
(EU project: ICT-FET OCTOPUS Integrating Project)

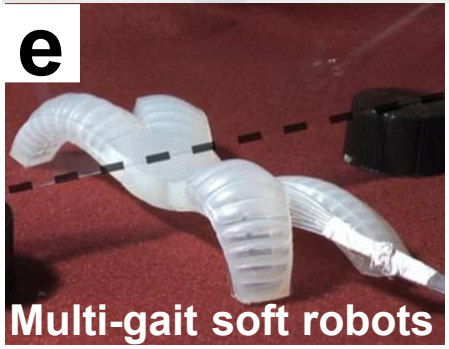
Soft robots

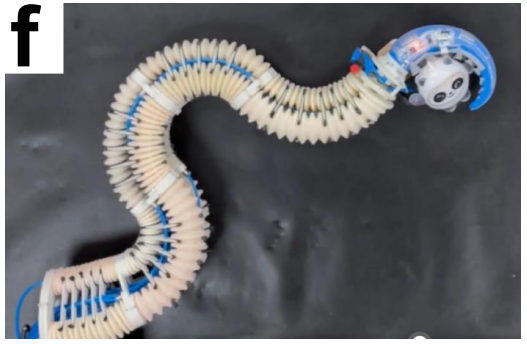










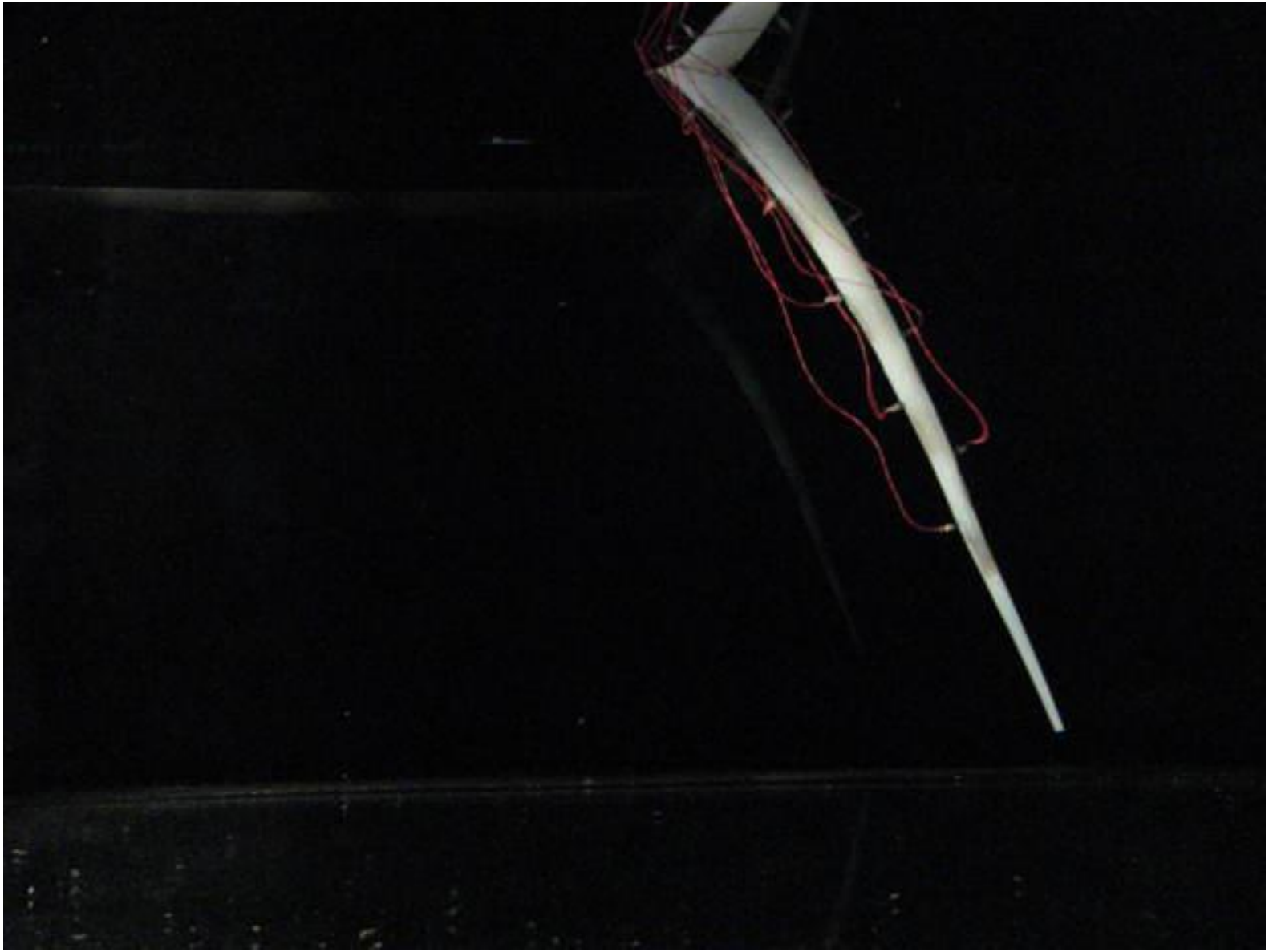


a. Cianchetti, M., et. al., Mater. Sci. Eng., C, 31, 1230-1239, 2011.. **b**. "Robotic Octopus Takes First Betentacled Steps", IEEE spectrum, 9 Apr 2012. **c.** Nature 536, 451–455 (2016) **d.** "Camouflage and Display for Soft Machines", Morin, S.A., et. al., Science, 2012, 337, 828-832. **e**. "Multi-Gait Soft Robot", Shepherd, R.F., et. al., PNAS, 108, 20400-20403, 2011. **f.** Sci. Robot.8,eadh7852(2023).

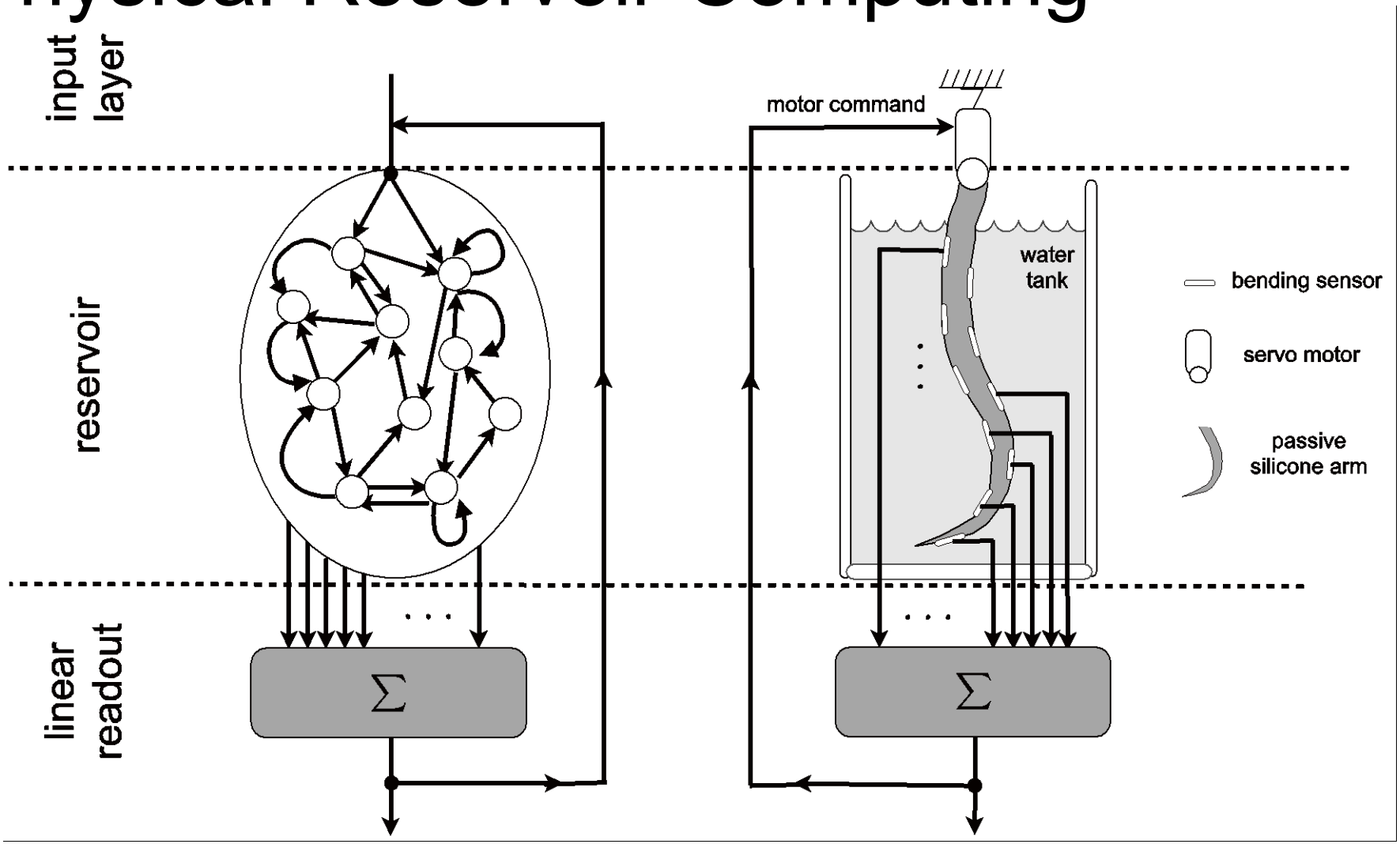
Good points

- Deformability
- Interactional safety

Computing with soft robotic arm?



Physical Reservoir Computing



K. Nakajima et al., *Front. Comput. Neurosci.*, 7 (91), 2013.
K. Nakajima et al., *J. R. Soc. Interface* 11: 20140437, 2014.
K. Nakajima et al., *Scientific Reports* 5: 10487, 2015.
K. Nakajima et al., *Soft Robotics 5(3): pp.339-347*, 2018.

Soft body dynamics as computational resource!

*Prerequisite: "*reproducible response*"

(target function)

$$y_{k+1} = g(u_k, u_{k-1}, \dots)$$

(reservoir computing)

$$\begin{cases} \boldsymbol{x_{k+1}} = f(\boldsymbol{x_k}, u_k) \\ \hat{y}_{k+1} = \hat{\psi}(\boldsymbol{x_{k+1}}) \end{cases}$$

$$g(u_k, u_{k-1}, \dots) \approx \hat{\psi}(\boldsymbol{x_{k+1}})$$

We want to emulate (learn) function "g"!

(prerequisite)

- Reproducible response to the same input sequence!
- Reservoir states should not depend on the initial condition!

$$f(\boldsymbol{x_k}, u_k) - f(\boldsymbol{x_k^*}, u_k) \approx 0 \quad \longleftarrow \quad \boldsymbol{x_k} = \boldsymbol{\phi}(u_{k-1}, u_{k-2}, \dots)$$

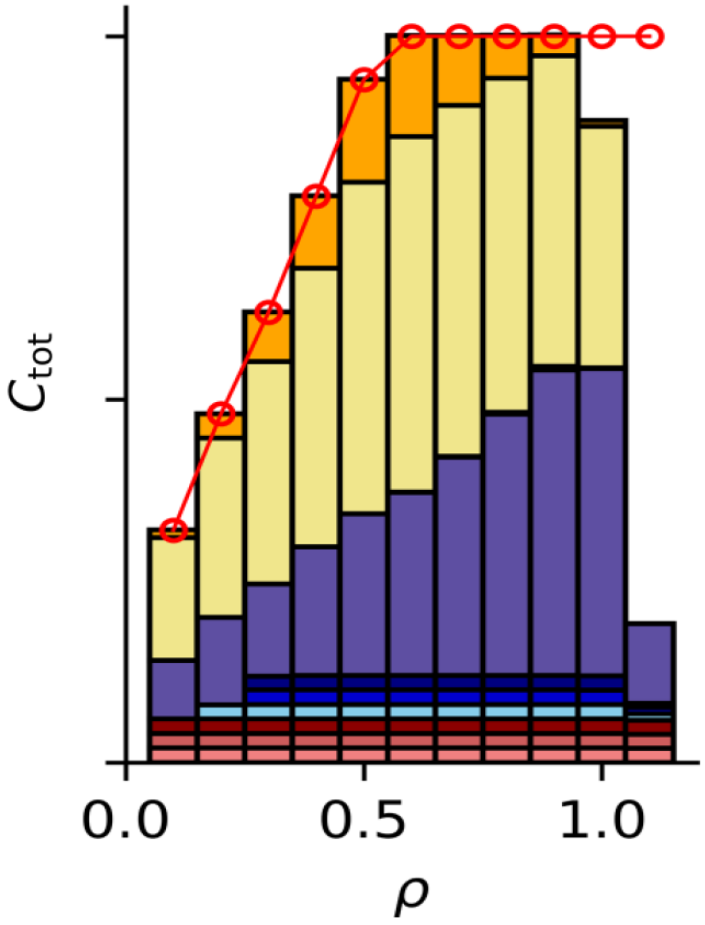
(common-signal-induced synchronization/ Negative conditional Lyapunov exponents)

Z. Lu, B. R. Hunt, E. Ott, *Chaos* 28, 061104 (2018).

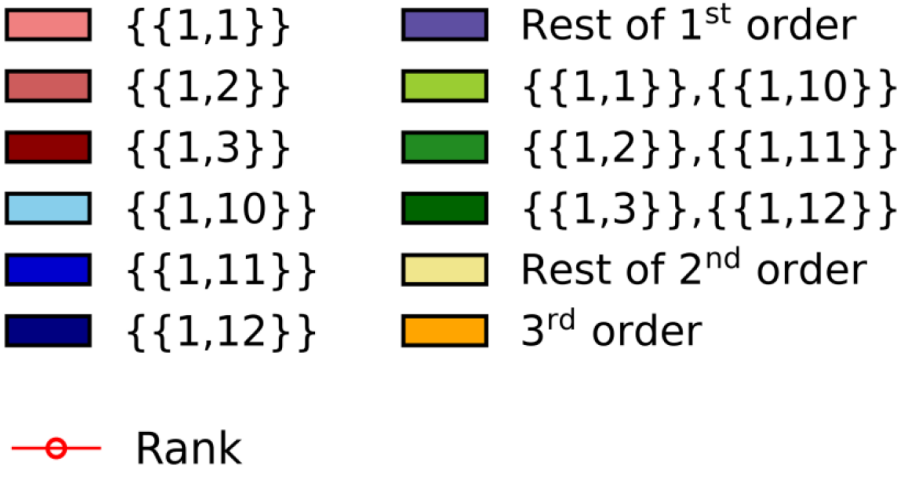
(echo state property (ESP))

H. Jaeger, GMD Technical Report. 148 (2001).
I. B. Yildiz, et al., *Neural netw.* 35 (2012).
G. Manjunath, et al., *Neural comp.* 25 (2013).

Information processing capacity in ESN



$$x_{i,t+1} = \tanh\left(\rho \sum_{j=1}^{N} w_{ij} x_{j,t} + \iota w_{\mathrm{in},i} \zeta_t\right)$$



Kubota, T., Takahashi, H., & Nakajima, K. (2021). Unifying framework for information processing in stochastically driven dynamical systems. *Physical Review Research*, *3*(4), 043135.

- Decompose $\boldsymbol{\phi}$ with products of orthogonal polynomials.
- Input range is [0, 1]. Legendre polynomial is used.

General solution of $x_{k+1} = f(x_k, u_k)$:

$$\boldsymbol{u}_k = \{u_{k-1}, u_{k-2}, \ldots, u_0\},\ \boldsymbol{x_k} = \boldsymbol{\phi}(\boldsymbol{u_k}, x_0, k)$$

- $\boldsymbol{x_k} = \boldsymbol{\phi}(\boldsymbol{u_k})$

Has ESP. It can be reliably used as reservoir!

- $\boldsymbol{x_k} = \boldsymbol{\phi}(\boldsymbol{u_k}, x_0, k)$

No ESP. Depends on both initial state and inputs.

- $\boldsymbol{x_k} = \boldsymbol{\phi}(\boldsymbol{u_k}, \boldsymbol{u'}_k, x_0, k)$

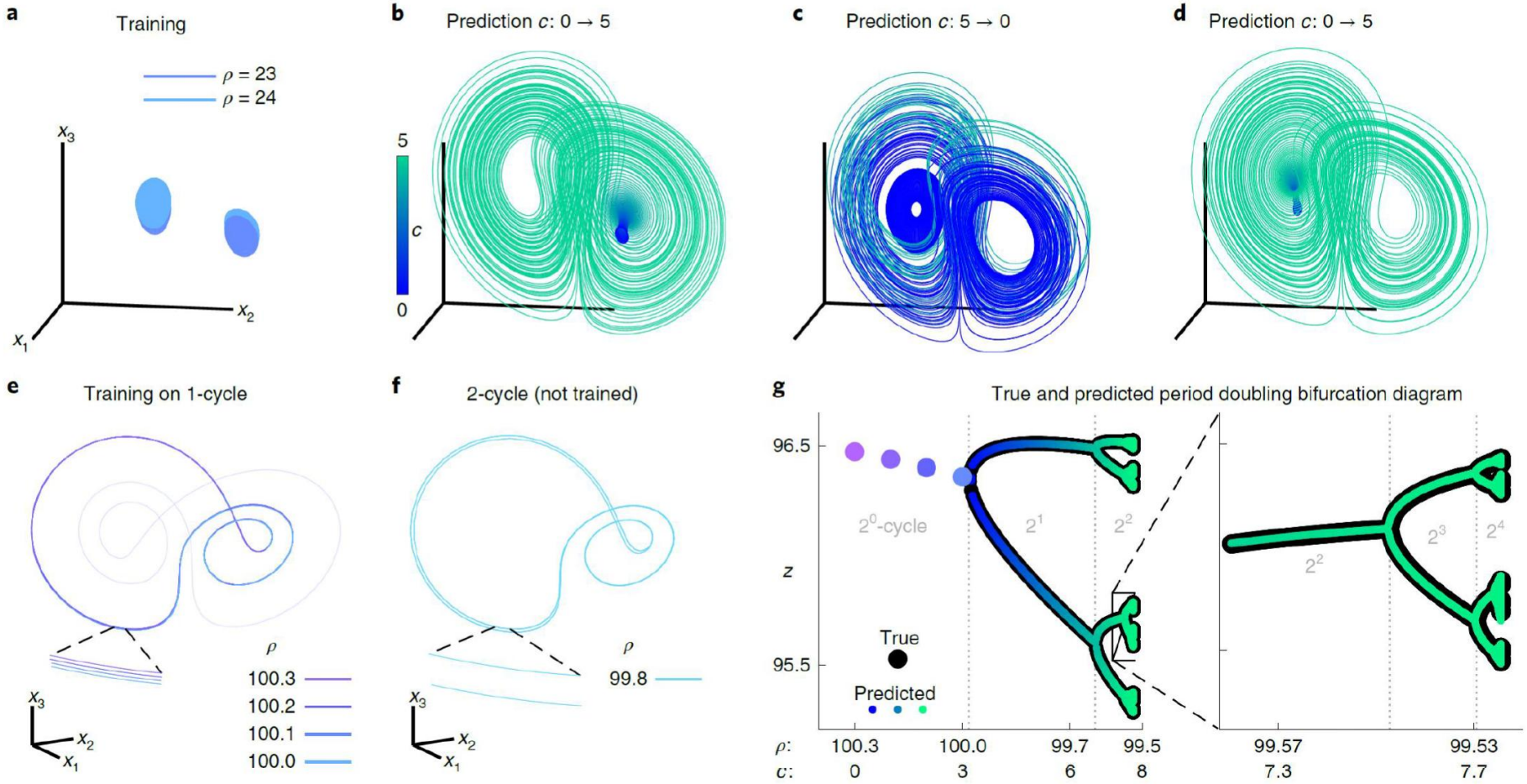
No ESP. Multiple inputs case, such as noise…

Are these dynamics useless for computation?
-> Recent hot topic!

Kubota, T., Imai, Y., Tsunegi, S., & Nakajima, K. (2024). Reservoir Computing Generalized. *arXiv preprint arXiv:2412.12104*.

Kubota, T., Takahashi, H., & Nakajima, K. (2021). Unifying framework for information processing in stochastically driven dynamical systems. *Physical Review Research*, *3*(4), 043135.

Learning bifurcations from limited examples through extrapolation



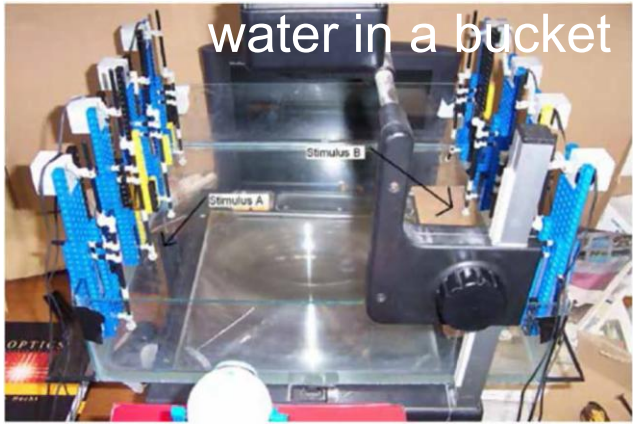
Kim, J. Z., Lu, Z., Nozari, E., Pappas, G. J., & Bassett, D. S. (2021). Teaching recurrent neural networks to infer global temporal structure from local examples. *Nature Machine Intelligence*, *3*(4), 316-323.

- Only by learning several attractors, the reservoir system can learn entire bifurcation structure!

Physical reservoirs …

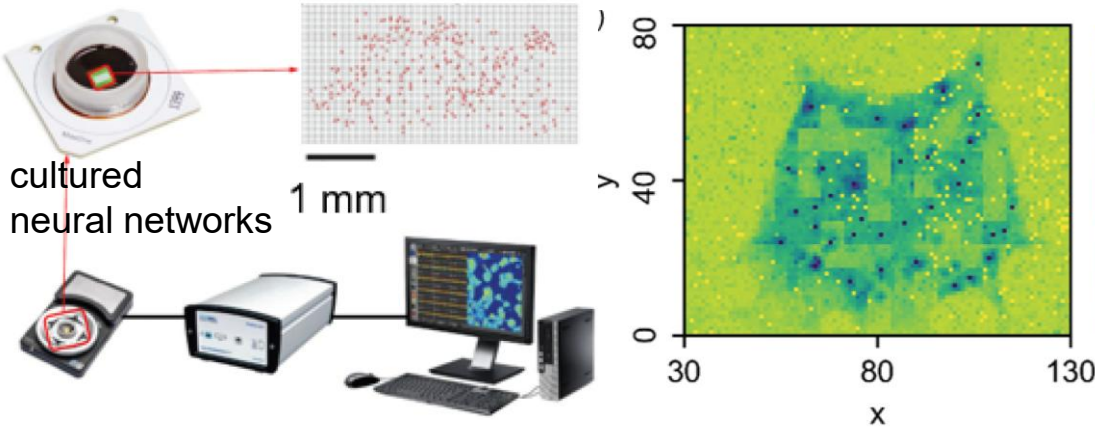
K. Nakajima, Physical reservoir computing---an introductory perspective, Jap. J. Appl. Phys. 59: 060501, 2020.

Liquid brain



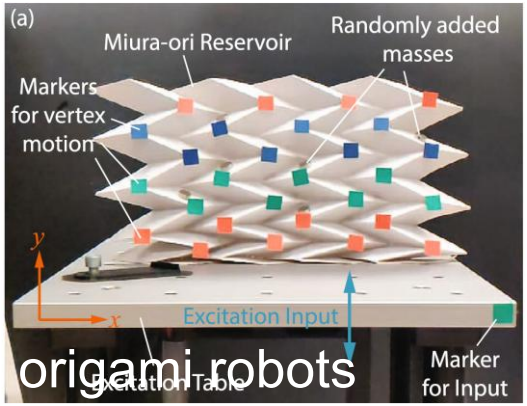
C. Fernando et al., Lec. Comp. Sci. 2801 (2003).

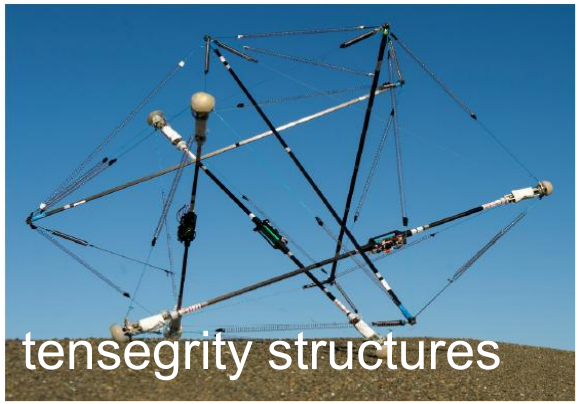
Cultured neural networks

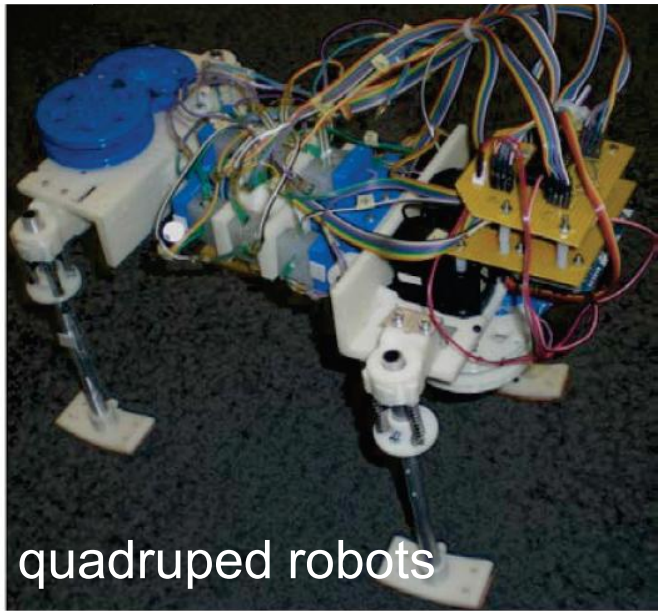


M. R. Dranias, et al., J. Neurosci. 33, 1940 (2013).
T. Kubota, et al., Lect. Comp. Sci. 11731 (2019).
Y. Yada, et al., Appl. Phys. Lett., 119(17), 173701 (2021).

Soft robots



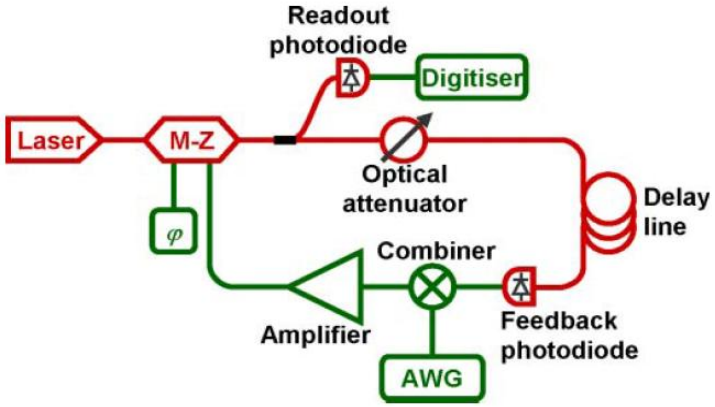




Q. Zhao et al., Proceedings of IROS, pp. 1445-1451 (2013).
K. Nakajima et al. J. R. Soc. Interface. 11: 20140437 (2014).
K. Caluwaerts et al. J. R. Soc. Interface 11:98 (2014).
K. Nakajima et al. Sci. Rep. 5: 10487 (2015).
K. Nakajima et al. Soft Robotics 5: 10487 (2018).
P. Bhovad, et al., Sci. Rep. 11(1), 1-18 (2021).

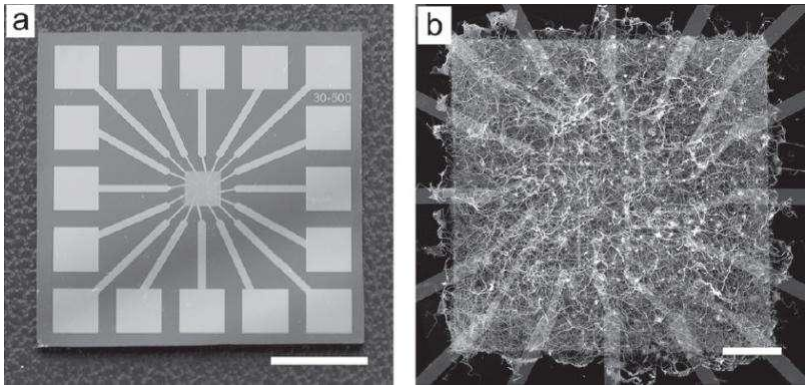
PRC for neuromorphic devices

Photonic reservoirs



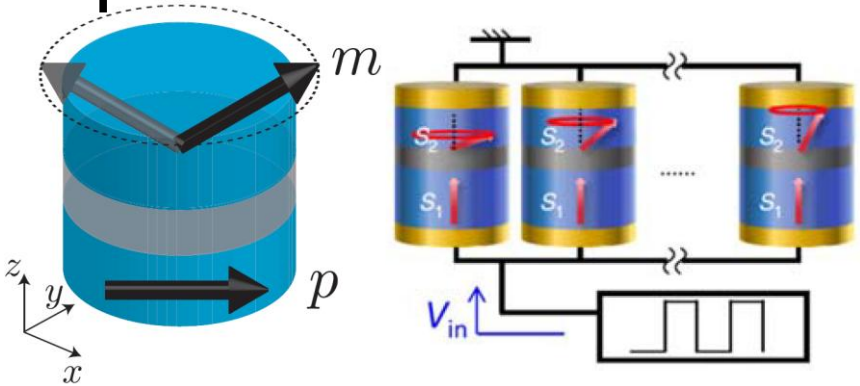
L. Larger, et al., Opt. Express 20, 3241 (2012).
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K. Vandoorne, et al., Nat. Commun. 5, 3541 (2014).
L. Larger, et al., Phys. Rev. X 7, 011015 (2017).
M. Nakajima et al.. Nat. Commun. 13, 7847 (2022).

In-Materia reservoirs



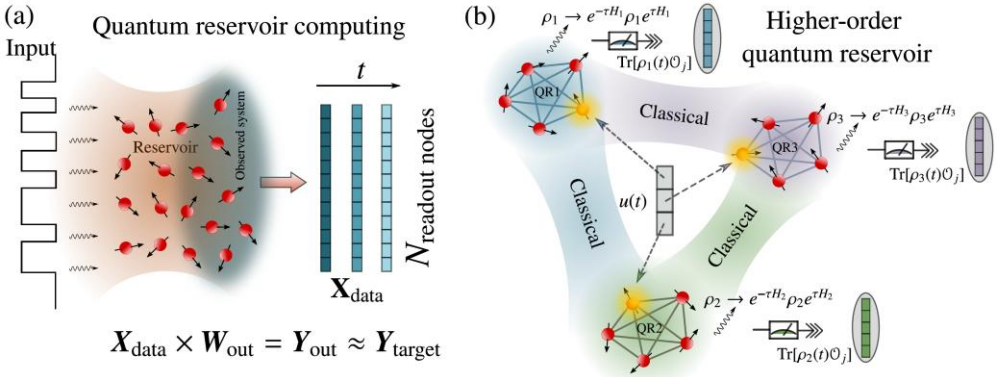
E. C. Demis et al. Nanotechnology 26:204003 (2015).
A. Z. Stieg et al. Adv. Mater. 24:286-293 (2012).
M. Cucchi, et al., Science Advances, 7(34), eabh0693 (2021).
Y. Usami, et al., Adv. Mater. (2021).

Spintronics reservoirs



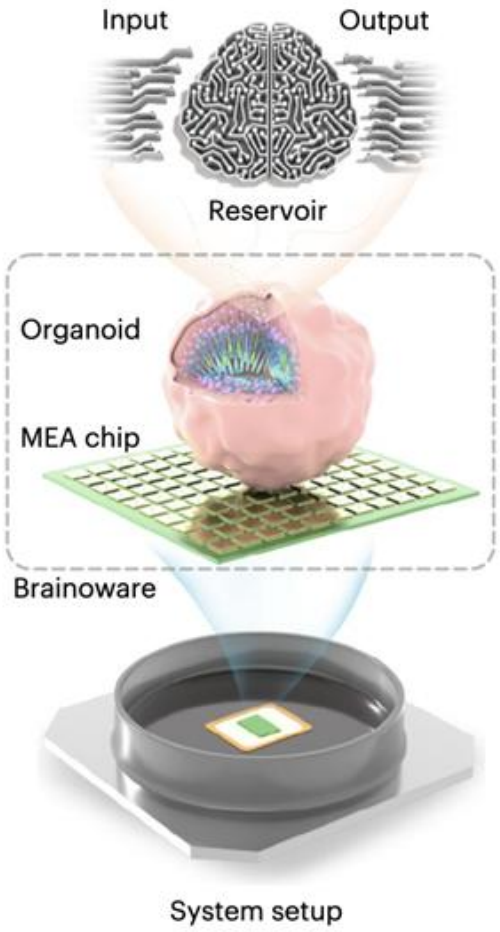
J. Torrejon et al., Nature 547, 428 (2017).
T. Furuta, et al., Phys. Rev. Appl. 10, 034063 (2018).
S. Tsunegi, et al., Appl. Phys. Lett. 114, 164101 (2019).
N. Akashi, et al., Phys. Rev. Res. 2: 043303 (2020).
Lee, O., Wei, T., Stenning, K.D. et al. Nat. Mater. (2023).

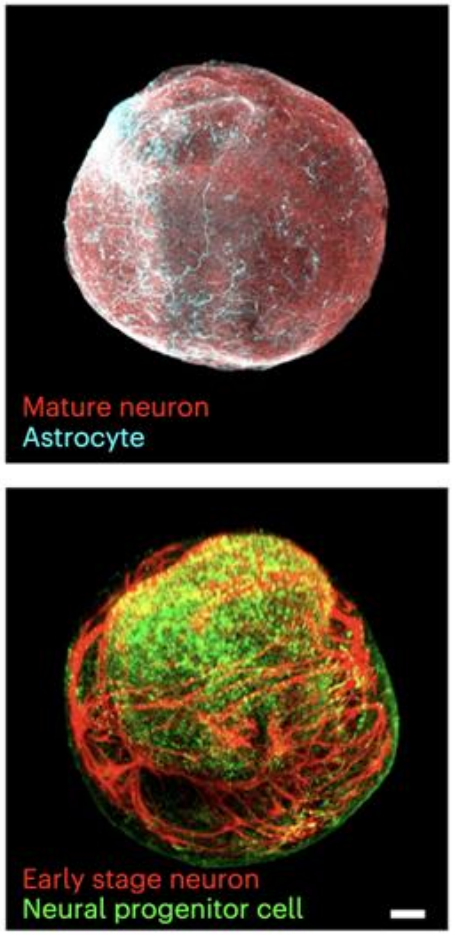
Quantum reservoirs



K. Fujii, K. Nakajima, Phys. Rev. Appl. 8: 024030 (2017).
K. Nakajima, et al., Phys. Rev. Appl. 11: 034021 (2019).
S. Ghosh, et al., Adv. Quantum Technol. 4: 2100053 (2021).
Q. H. Tran, K. Nakajima, Phys. Rev. Lett. 127: 260401 (2021).
T. Kubota et al., Phys. Rev. Res., 5(2), 023057 (2023)..

Brain Organoid Reservoir Computing



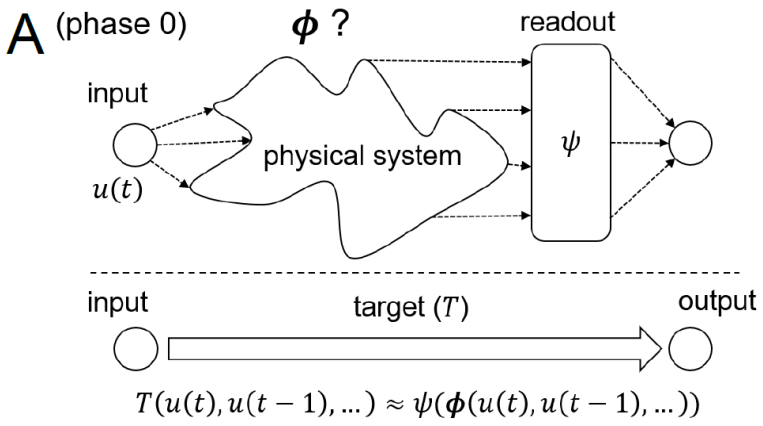


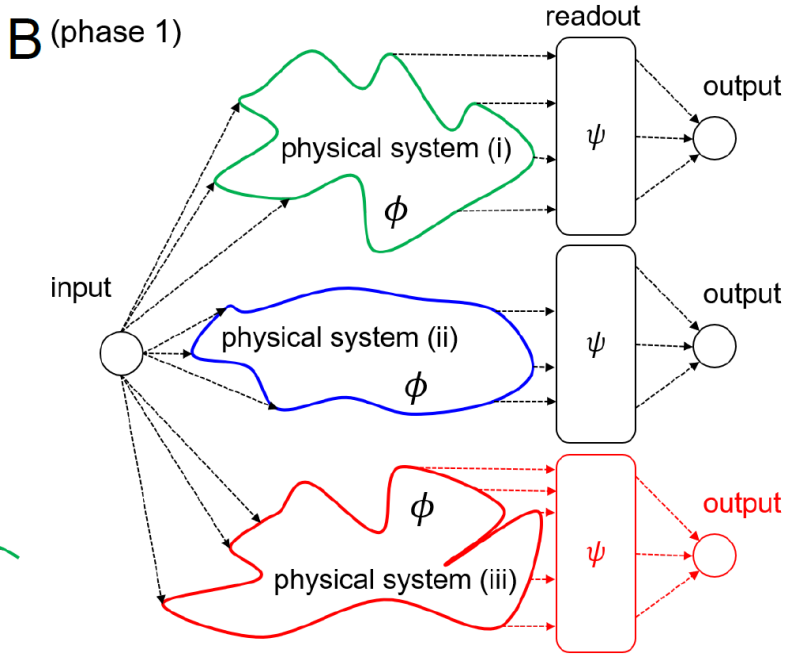
…We illustrate the practical potential of this technique by using it for speech recognition and nonlinear equation prediction in a ***reservoir computing framework***.

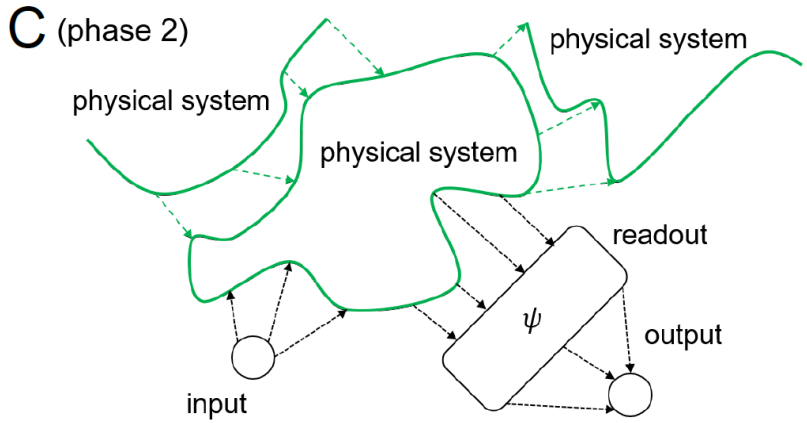
Cai, H., Ao, Z., Tian, C. *et al.* Brain organoid reservoir computing for artificial intelligence. *Nat Electron* (2023). https://doi.org/10.1038/s41928-023-01069-w

What is specific in PRC?

Nakajima, K. (2020). Physical reservoir computing—an introductory perspective. Jap. J. Appl. Phys., 59(6), 060501.







physical systems: function + its property

(i): ϕ + "extremely fast processing"

(ii): ϕ + "robust to radiation"

(iii): ϕ + "applicable to quantum tasks"

A: Inferring the computational power of physical systems.

B: Physical properties of a computer.

C: Exploiting a physical substrate that is not made for computation for computation.

A unicellular organism as a reservoir!

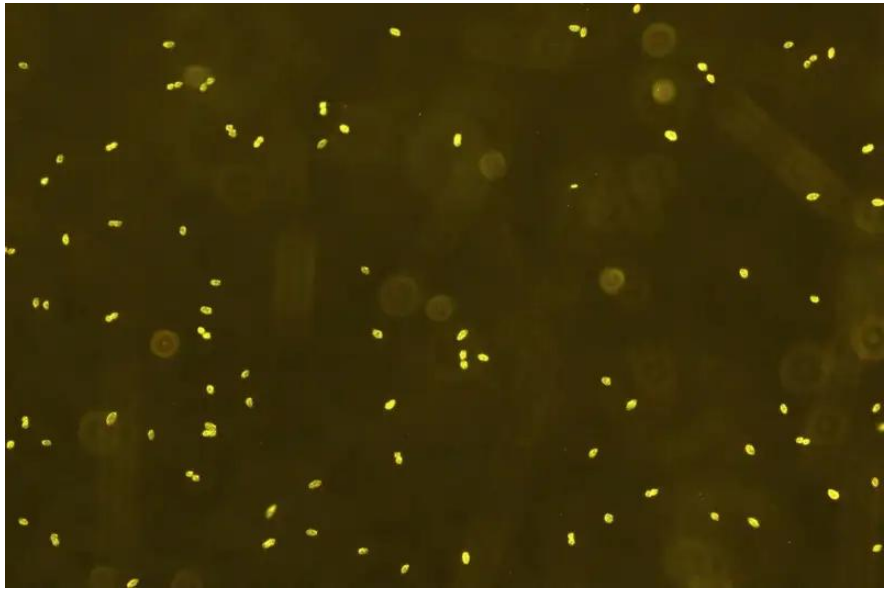
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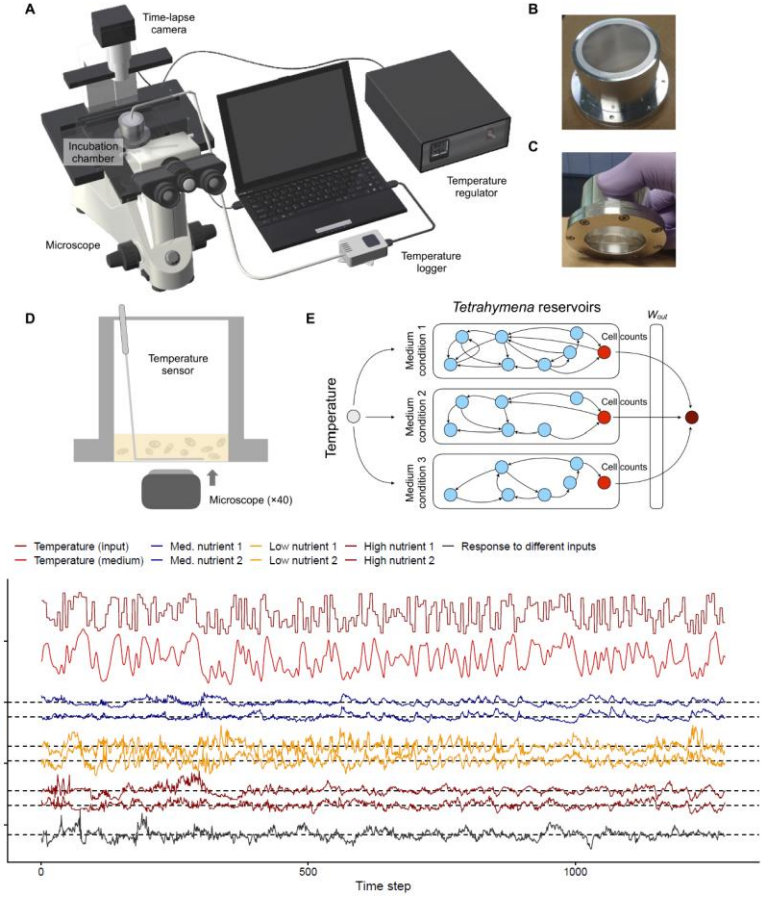
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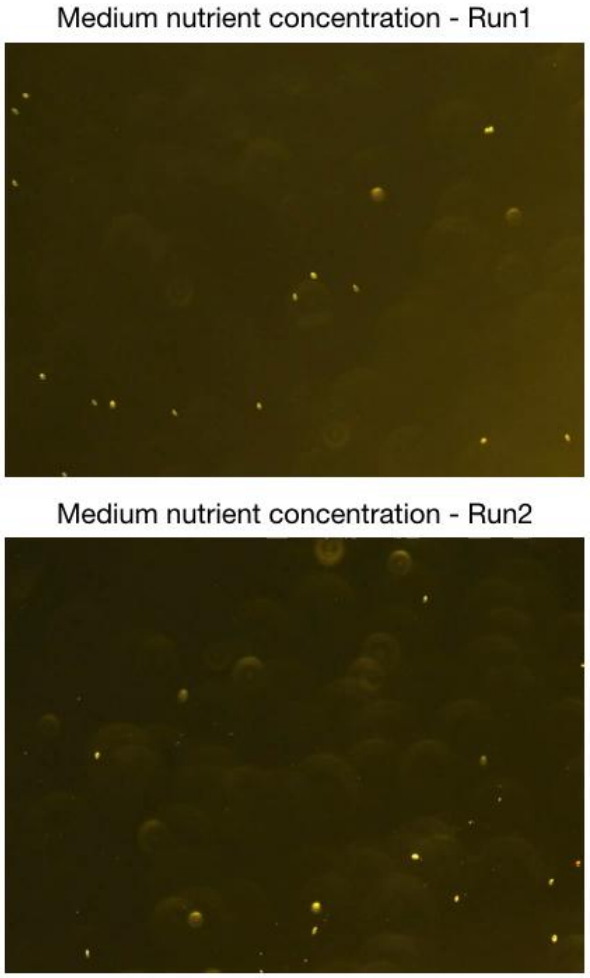
Biological computer made from single-celled organisms can crunch data

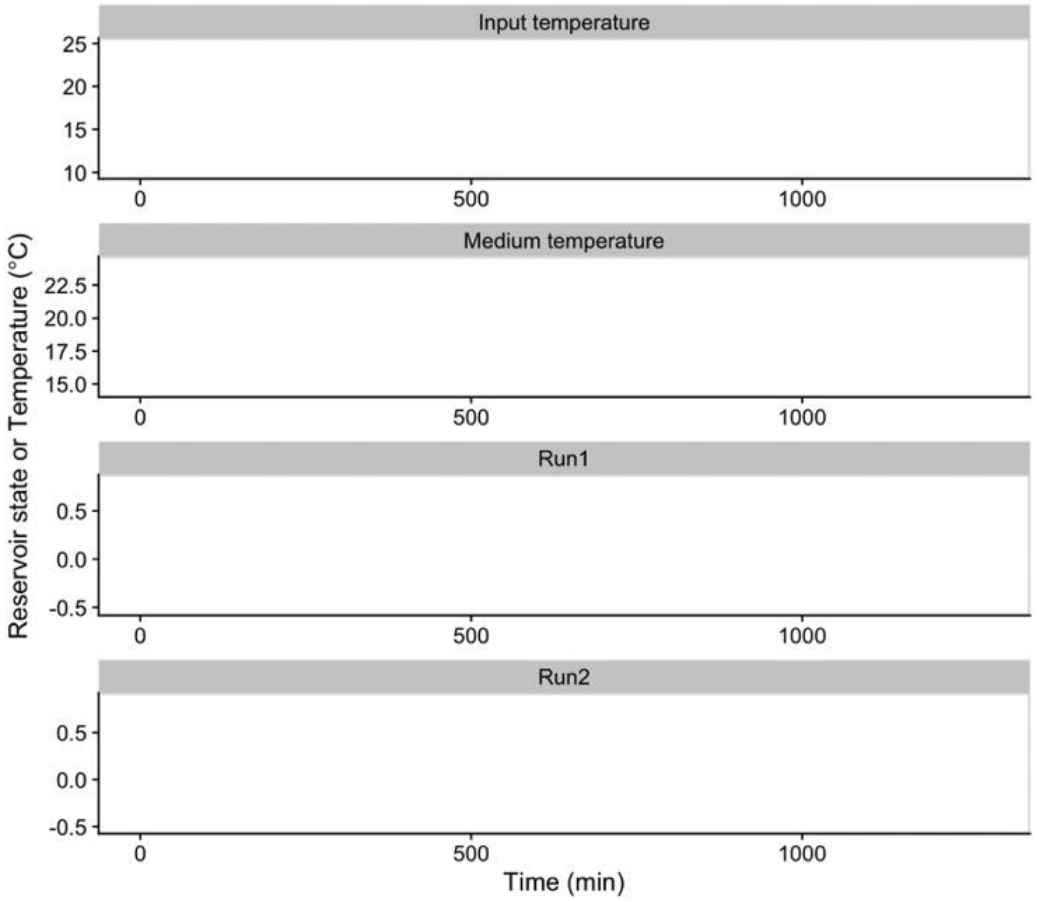




M. Ushio, K. Watanabe, Y. Fukuda, Y. Tokudome, K. Nakajima, Computational capability of ecological dynamics, *Royal Society Open Science* 10: 221614, 2023.

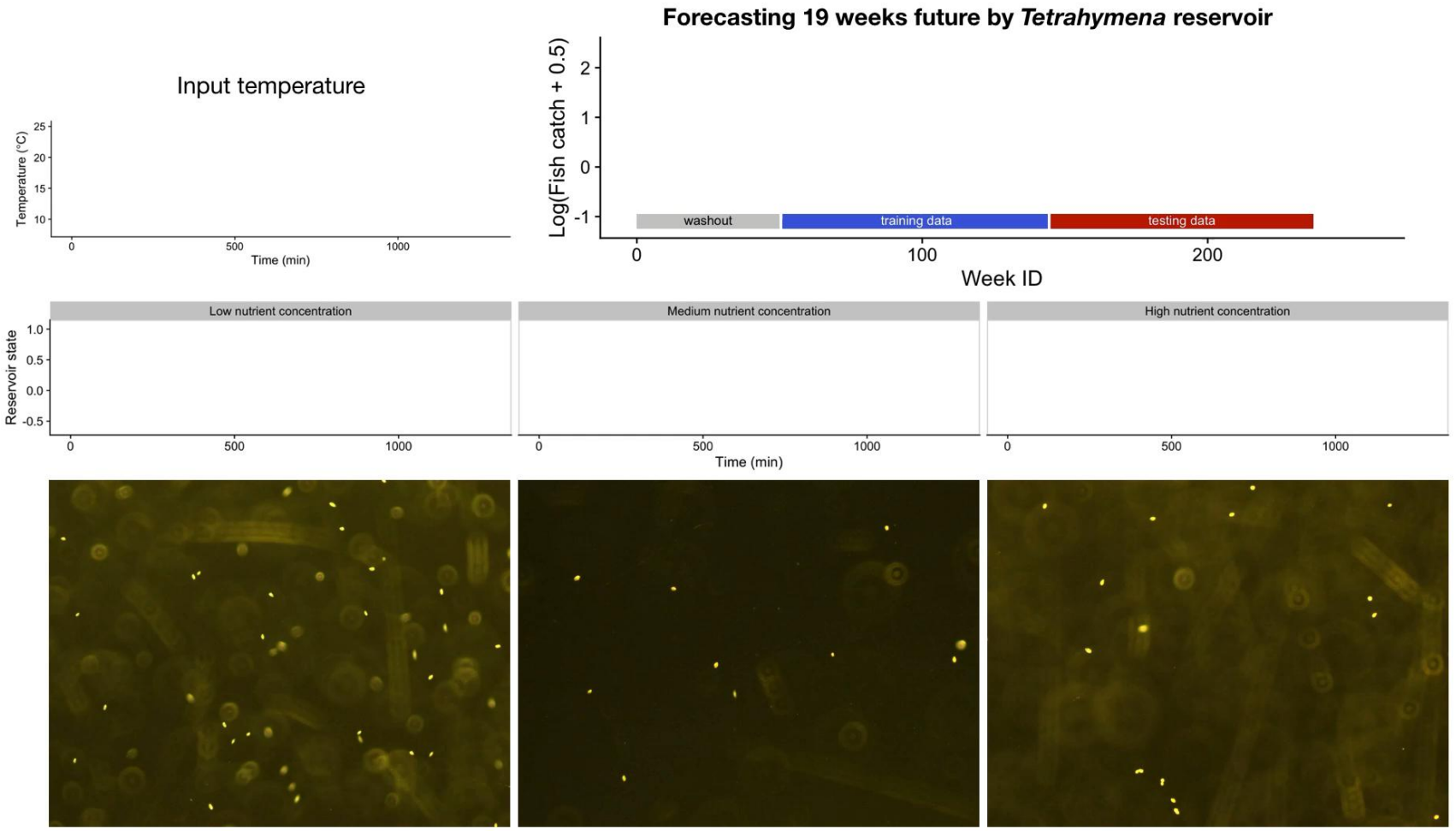
Common-signal-induced synchronization in ecological dynamics: *Tetrahymena thermophila*





M. Ushio, K. Watanabe, Y. Fukuda, Y. Tokudome, K. Nakajima, Computational capability of ecological dynamics, Royal Society Open Science 10: 221614, 2023.

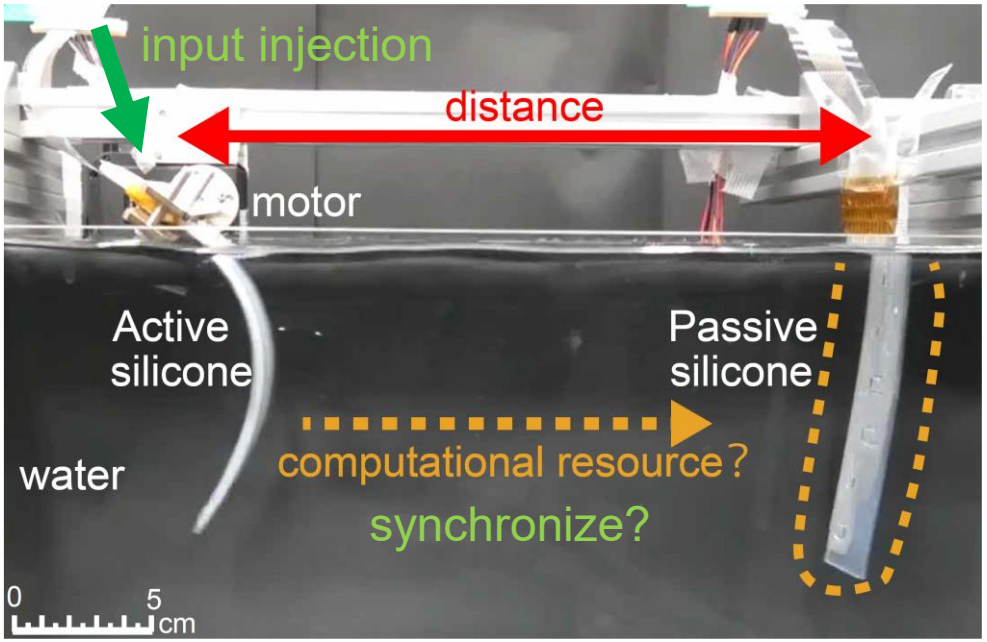
Forecasting population dynamics of fish using *Tetorahymena*



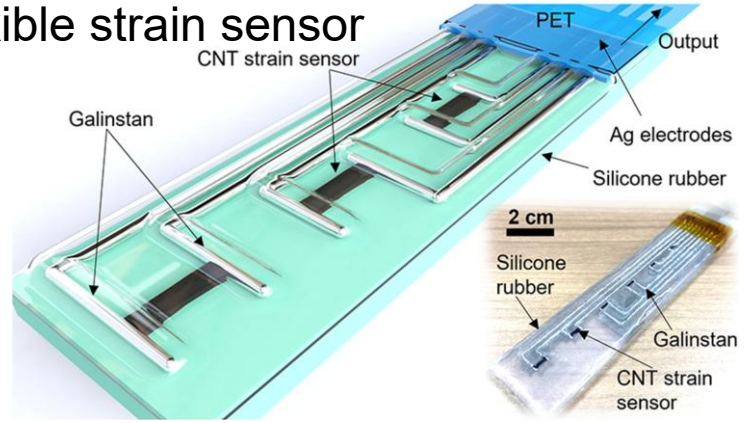
M. Ushio, K. Watanabe, Y. Fukuda, Y. Tokudome, K. Nakajima, Computational capability of ecological dynamics, Royal Society Open Science 10: 221614, 2023.

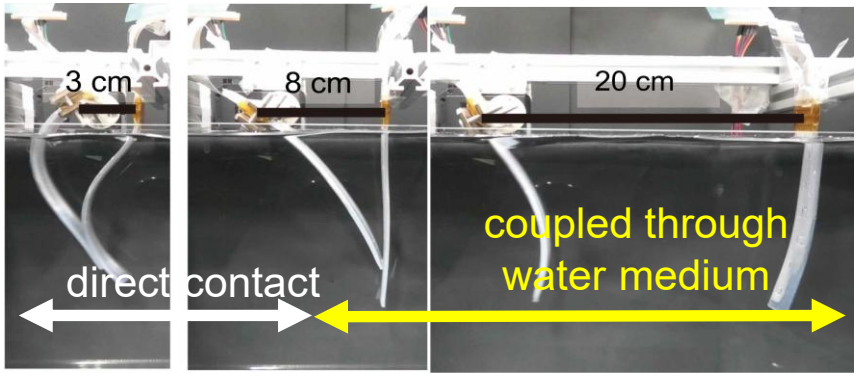
Remote reservoirs

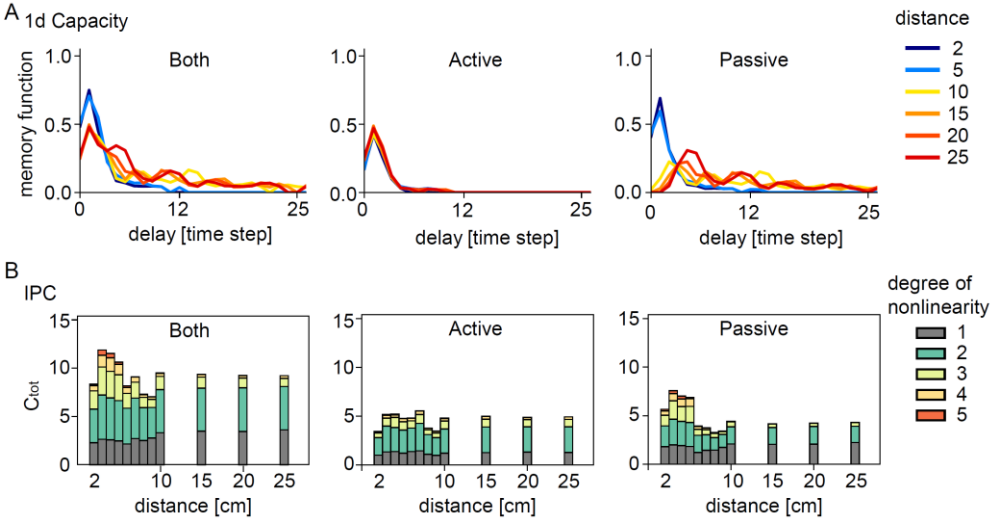
K. Tanaka, et al., Adv. Intell. Syst. 2100166, 2022.



Flexible strain sensor



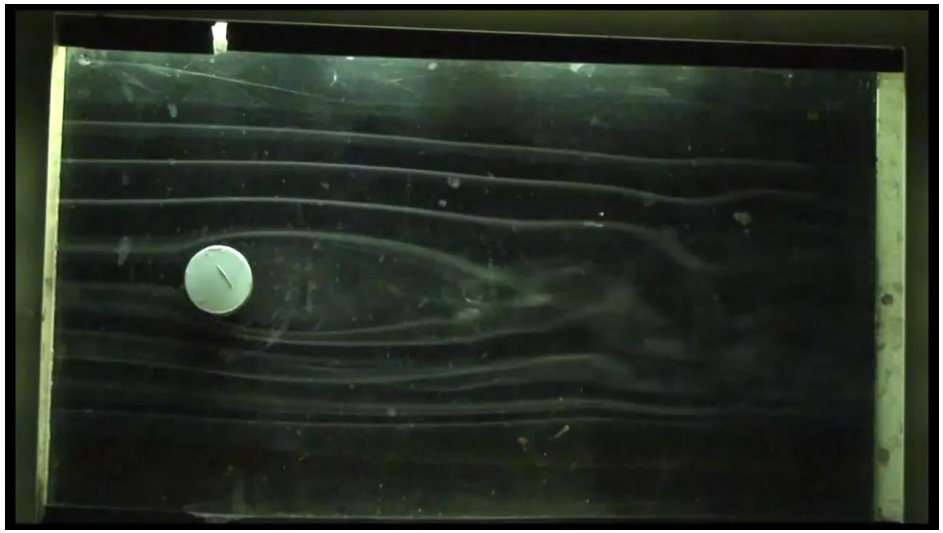


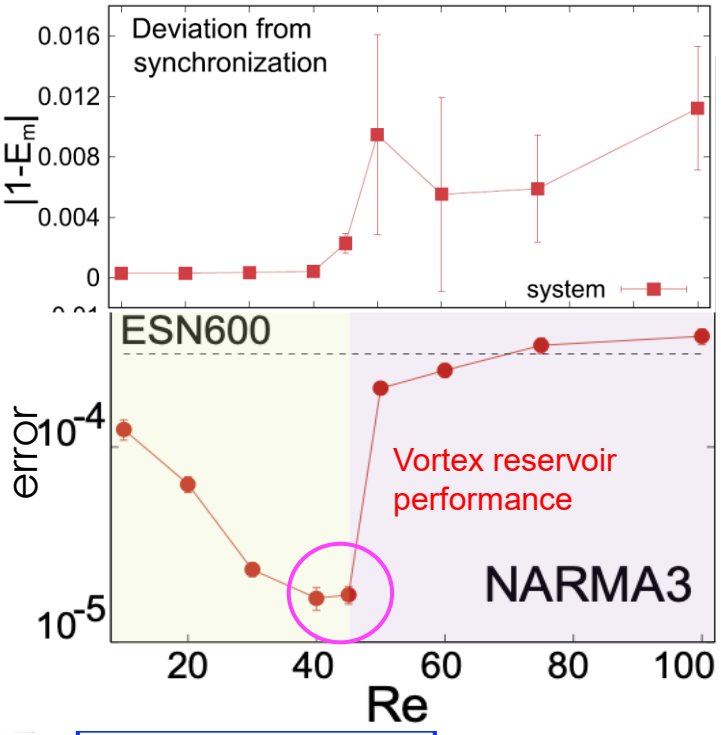


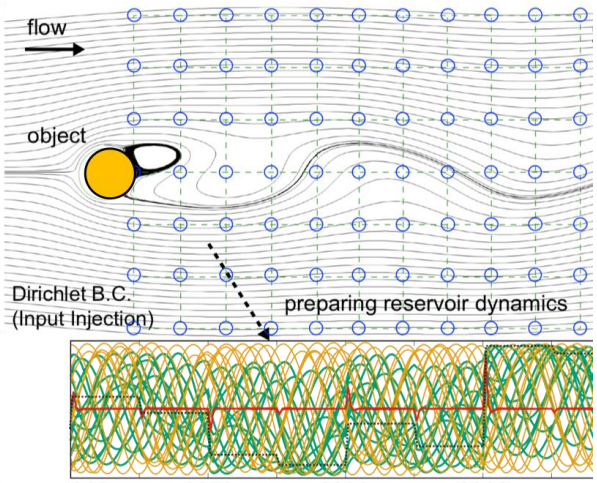
T. Kubota, et al., Phys. Rev. Res. 3: 043135, 2021.

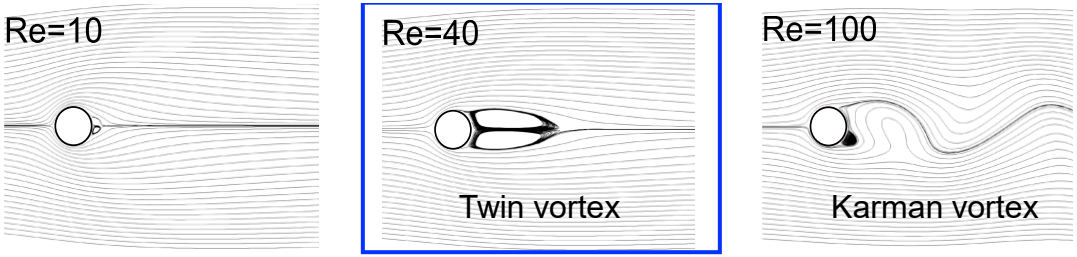
- Physical objects in spatially distance place can be used as a reservoir!
- Generalized synchronization induces self-organized *remote reservoirs*! (extending body)
- Interaction modality changes the computational capability!

Information processing in Karman vortex system





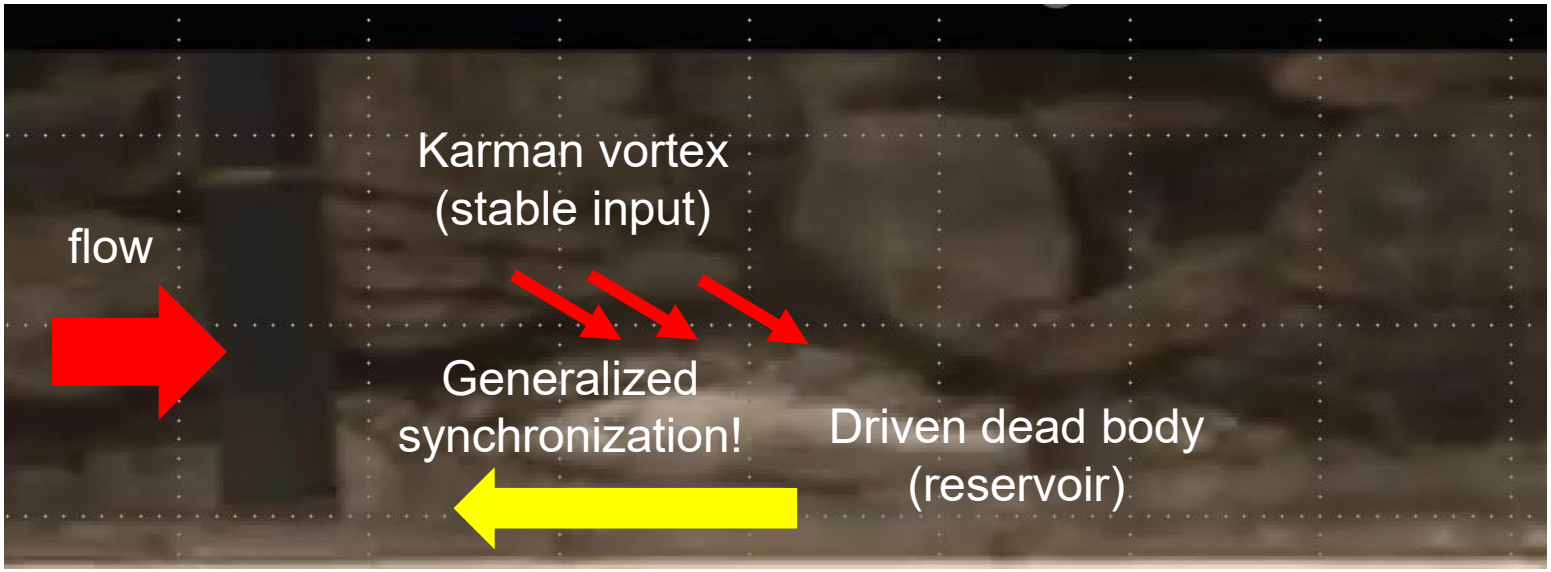




K. Goto, K. Nakajima, H. Notsu, Twin vortex computer in fluid flow, New Journal of Physics 23: 063051, 2021.

- Information processing capability is maximized at the break down of twin vortex (bifurcation point) -> Karman vortex cannot be used as a successful reservoir…
- However, it is robust as an input stream!

Understanding dead trout experiment as information processing



- Karman vortex (no ESP) as a robust and stable input against noise. (input embedded in nature)
- Generalized synchronization with Karman vortex (input) and dead body (reservoir).

Statement:

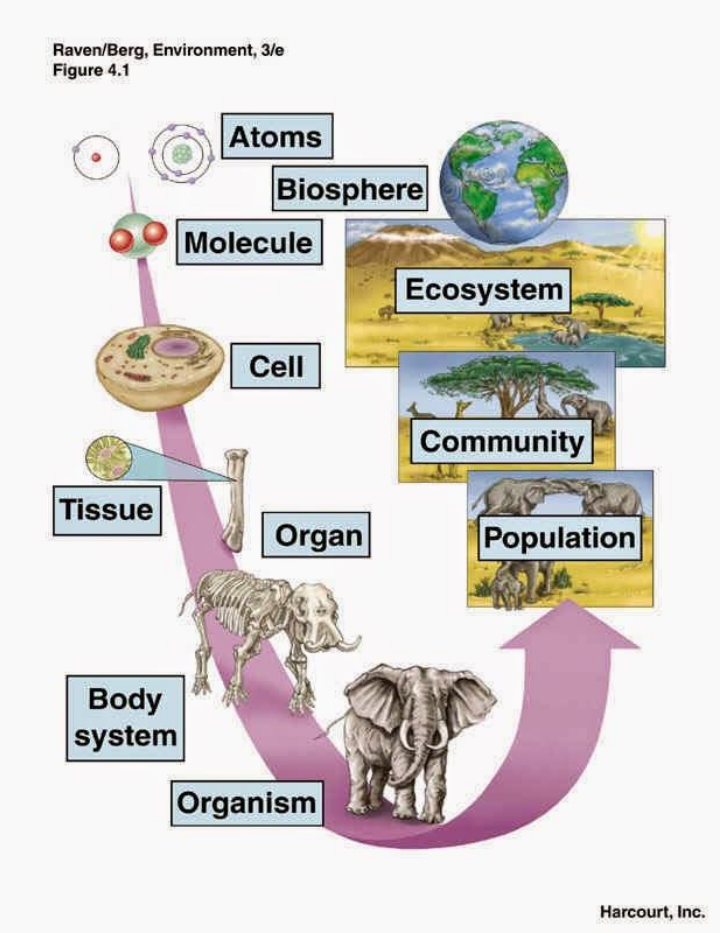
“By incorporating soft materials, the body can be used as a computational device!”

- Distributed computation – open-loop
- Embedding motor program into the body – closed-loop

Body structure of robots (animals) is not “random” but has specific morphology.

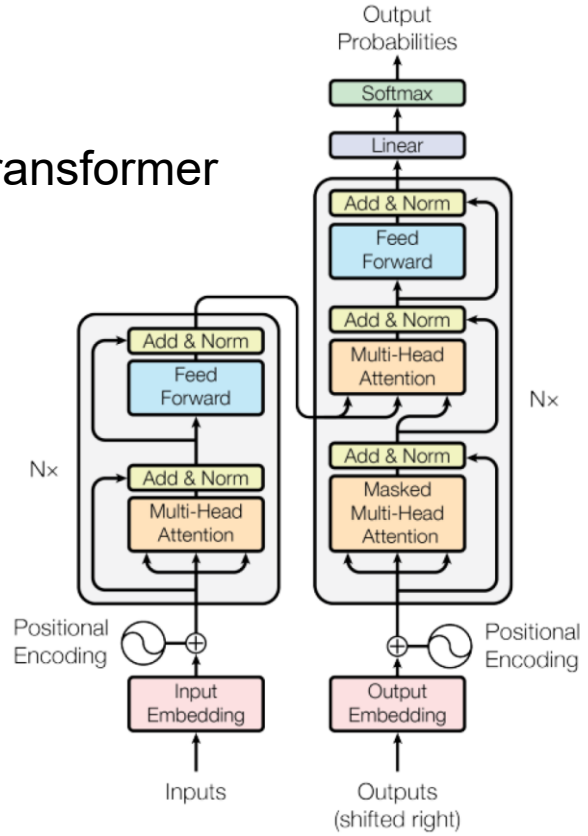
→ Explore “why” from computational perspectives!

Natural systems vs Large Language Models



- Evolved systems according to their ecological niche

Transformer

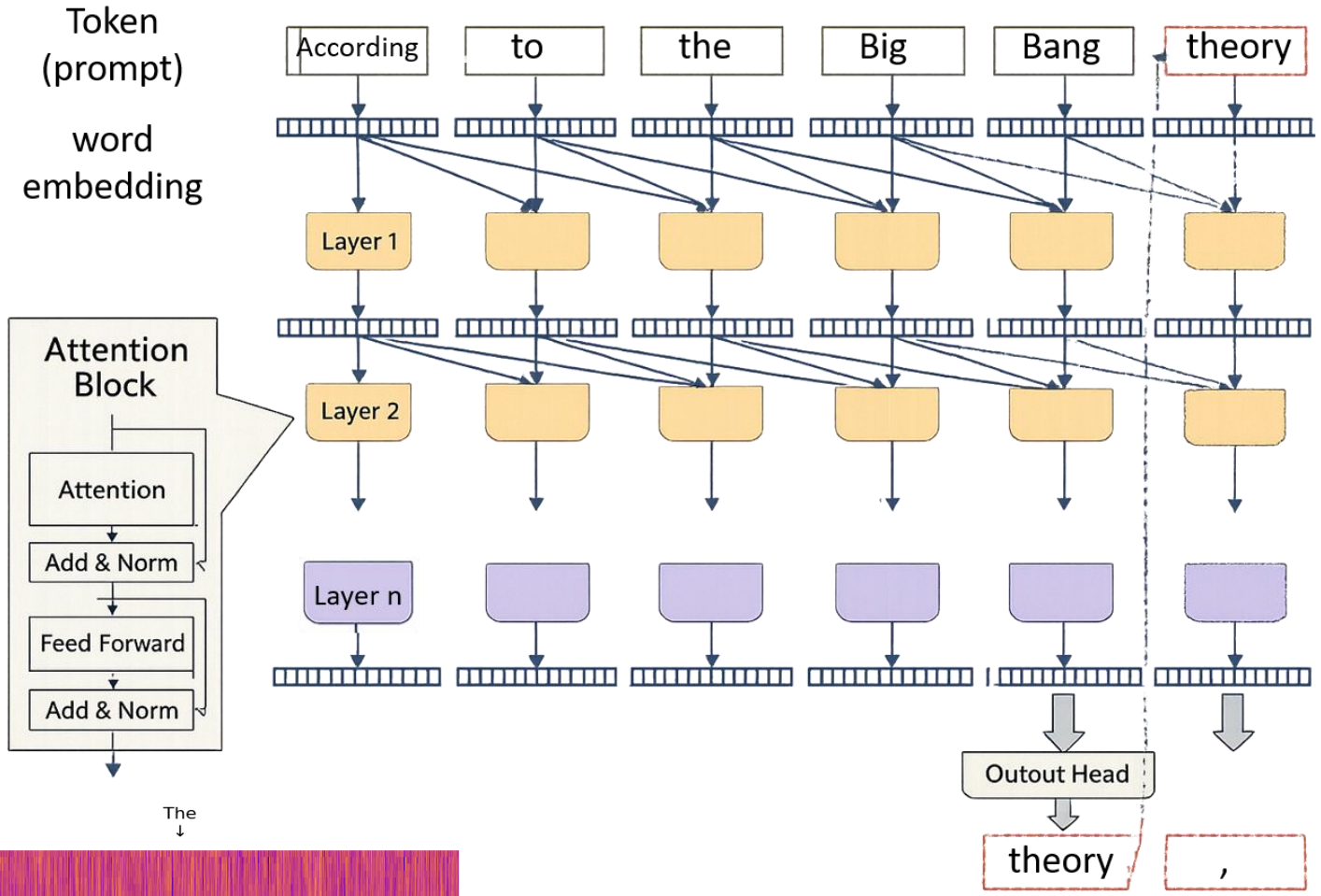


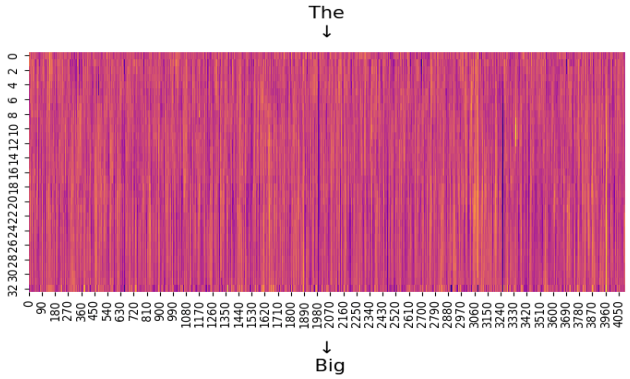
[1] Vaswani, A., et al., (2017). Attention is all you need. Advances in neural information processing systems, 30. Image from https://proceedings.neurips.cc/paper/2017/file/3f5ee243547dee91fbd053c1c4a845aa-Paper.pdf

- Trained through huge amount of sentence data (e.g. BERT: masked language modeling task)

Large language models as a reservoir

What is happening inside LLM?

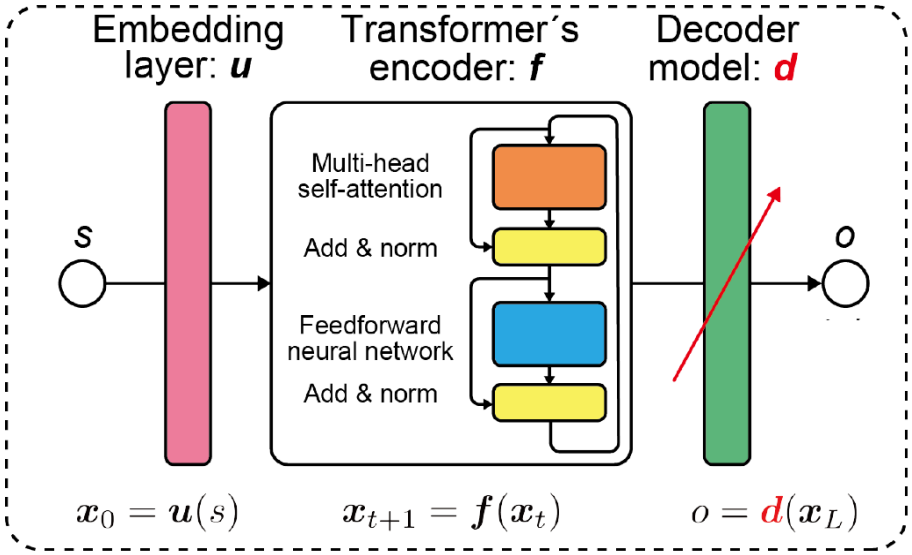


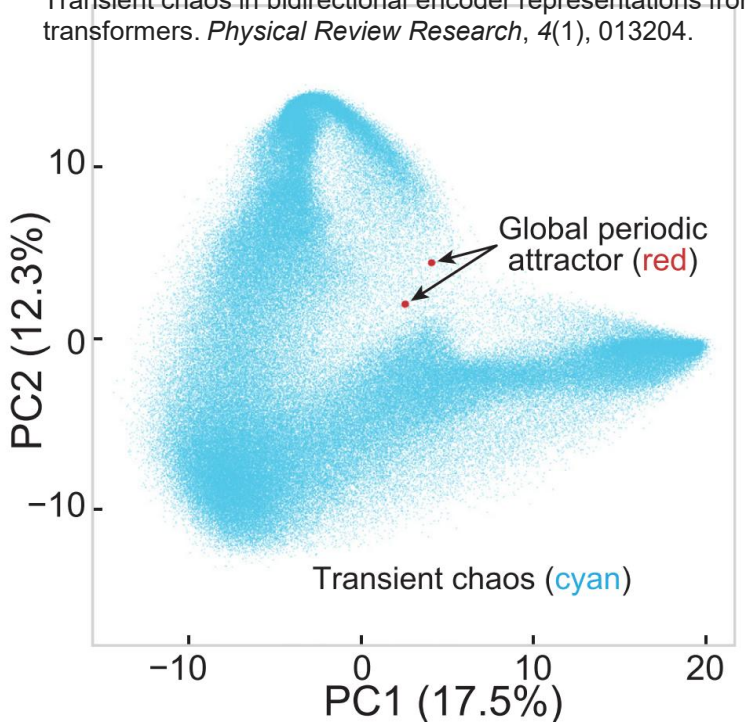


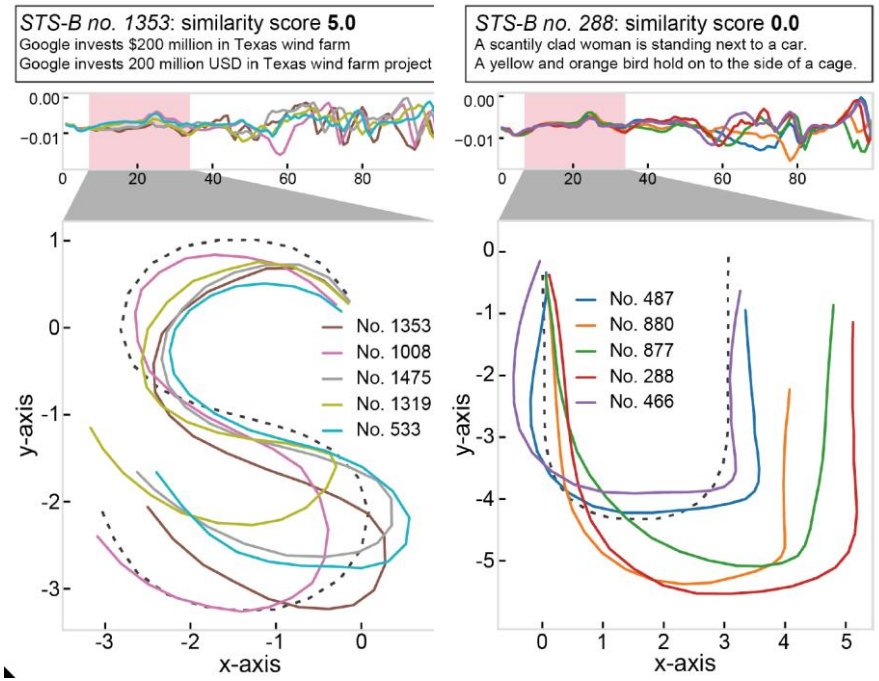
- Complex dynamics exists inside LLM!
- Can consider as a reservoir!

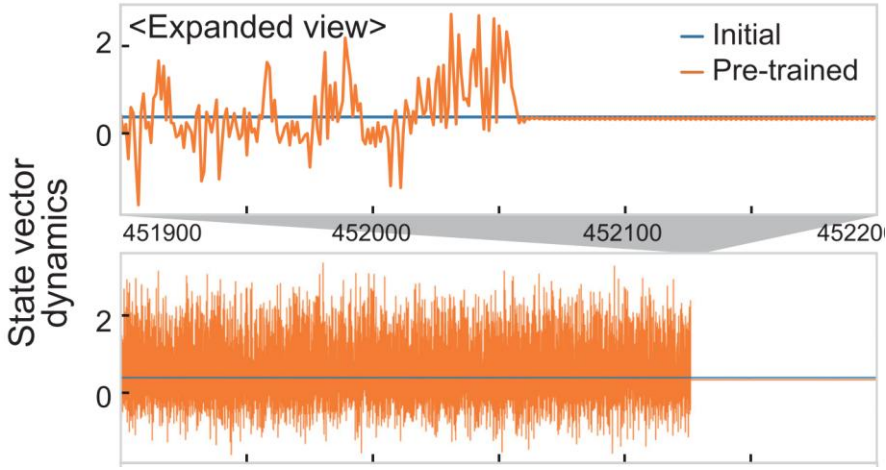
“Butterfly”(chaos) in BERT

Inoue, K., Ohara, S., Kuniyoshi, Y., & Nakajima, K. (2022). Transient chaos in bidirectional encoder representations from transformers. *Physical Review Research*, *4*(1), 013204.









- Transient chaos exists!

PRESS RELEASES

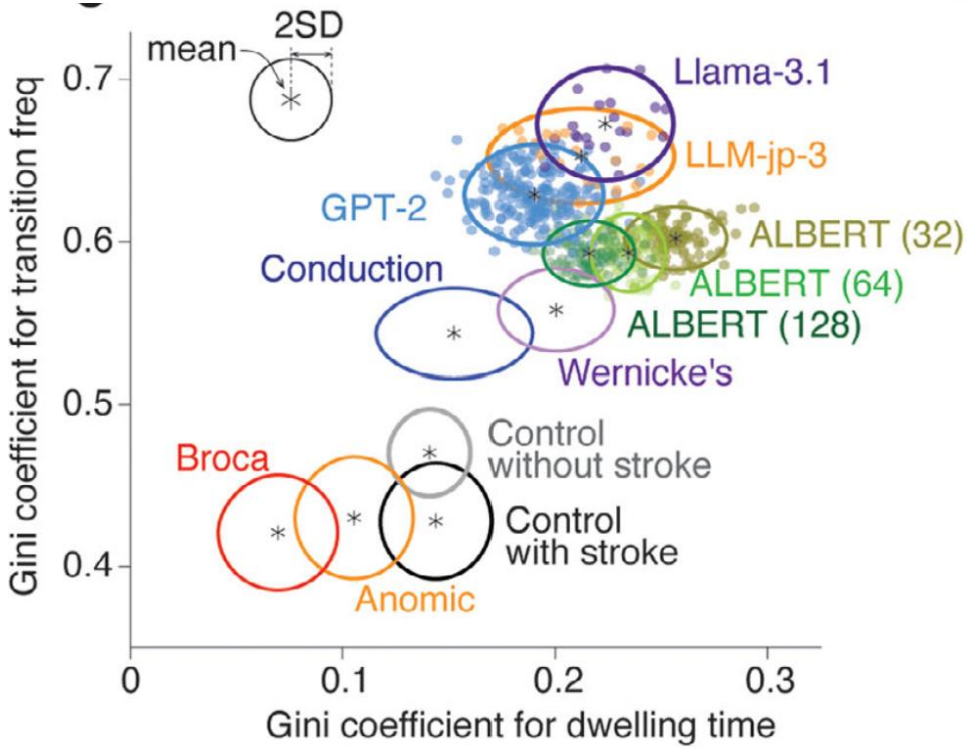
Comparing LLM with human brain dynamics

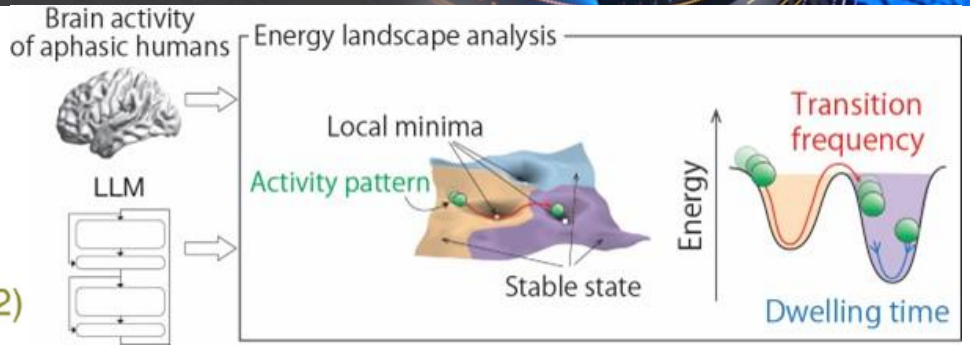
AI overconfidence mirrors human brain condition

A similarity between language models and aphasia points to diagnoses for both

Research news



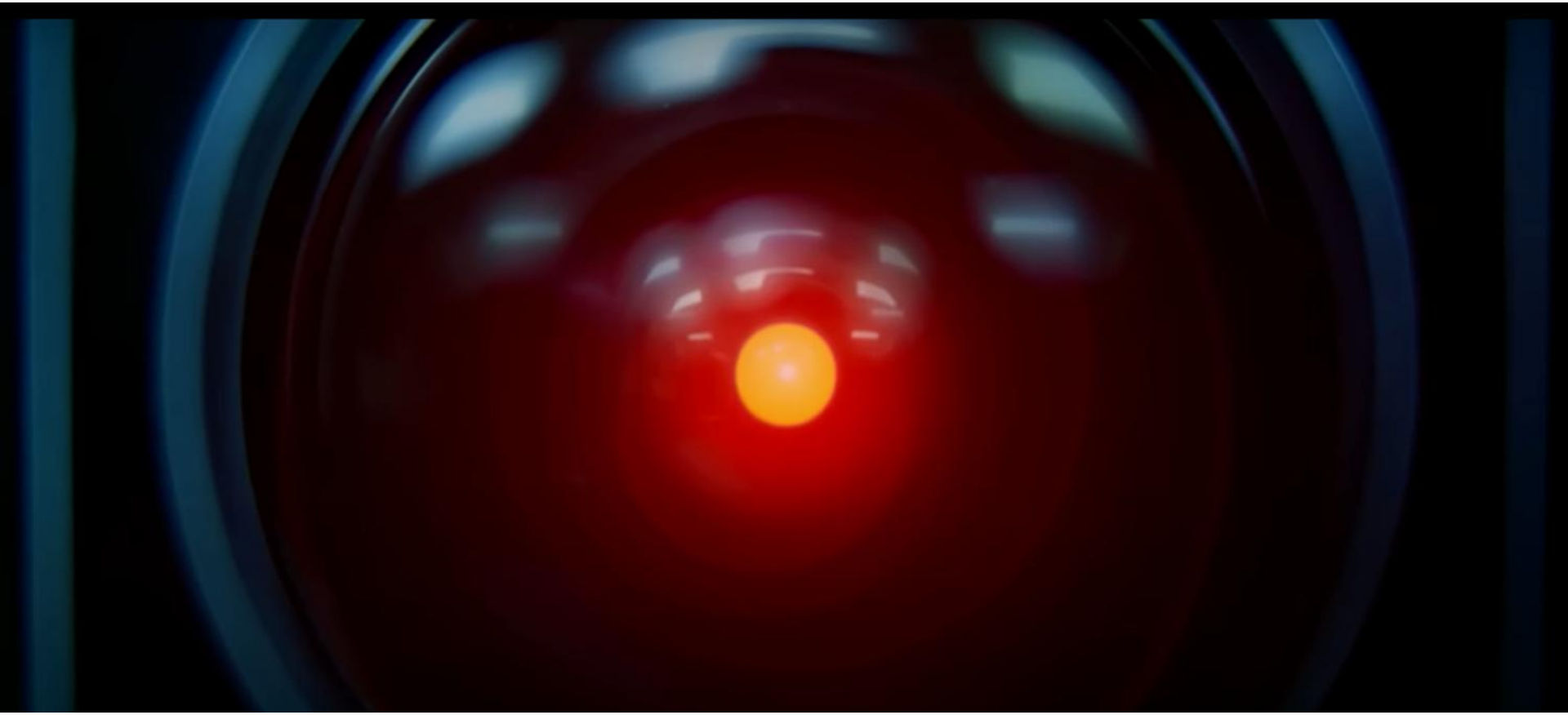




- Energy landscape analysis in the brain activity of aphasia patients and the internal dynamics of LLM
- LLM resembles sensory aphasia, as exemplified by Wernicke's aphasia!

T. Watanabe, K. Inoue, Y. Kuniyoshi, K. Nakajima, K. Aihara, Comparison of large language model with aphasia, ***Advanced Science***, 2414016, 2025.

“Hal9000” -- 2001: A Space Odyssey (1968)



“I'm Sorry, Dave, I can’t…”

Embodiment: *two bodies, two timescales*

- **Functional layer**

sensorimotor coupling, brain-body-environment

$$f$$

- **Constitutional layer**

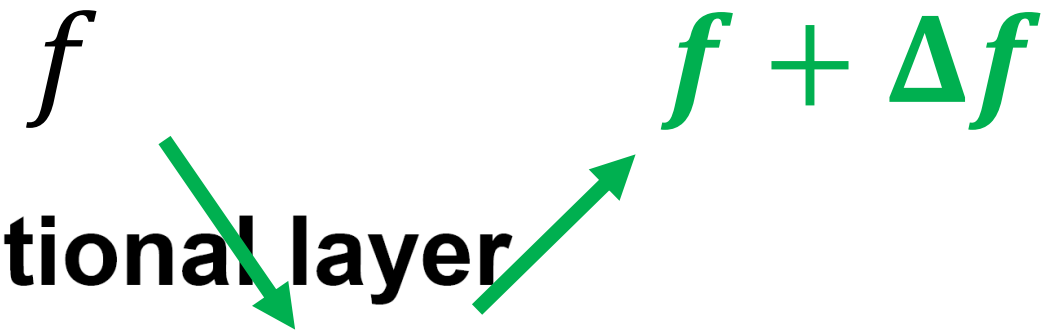
physical material that constitutes a function, evolvable hardware, developmental aspect, edible and degradable, metabolism and energy

Interaction between **functional** and its **constitutional layers**!

Embodiment: *two bodies, two timescales*

- **Functional layer**

sensorimotor coupling, brain-body-environment

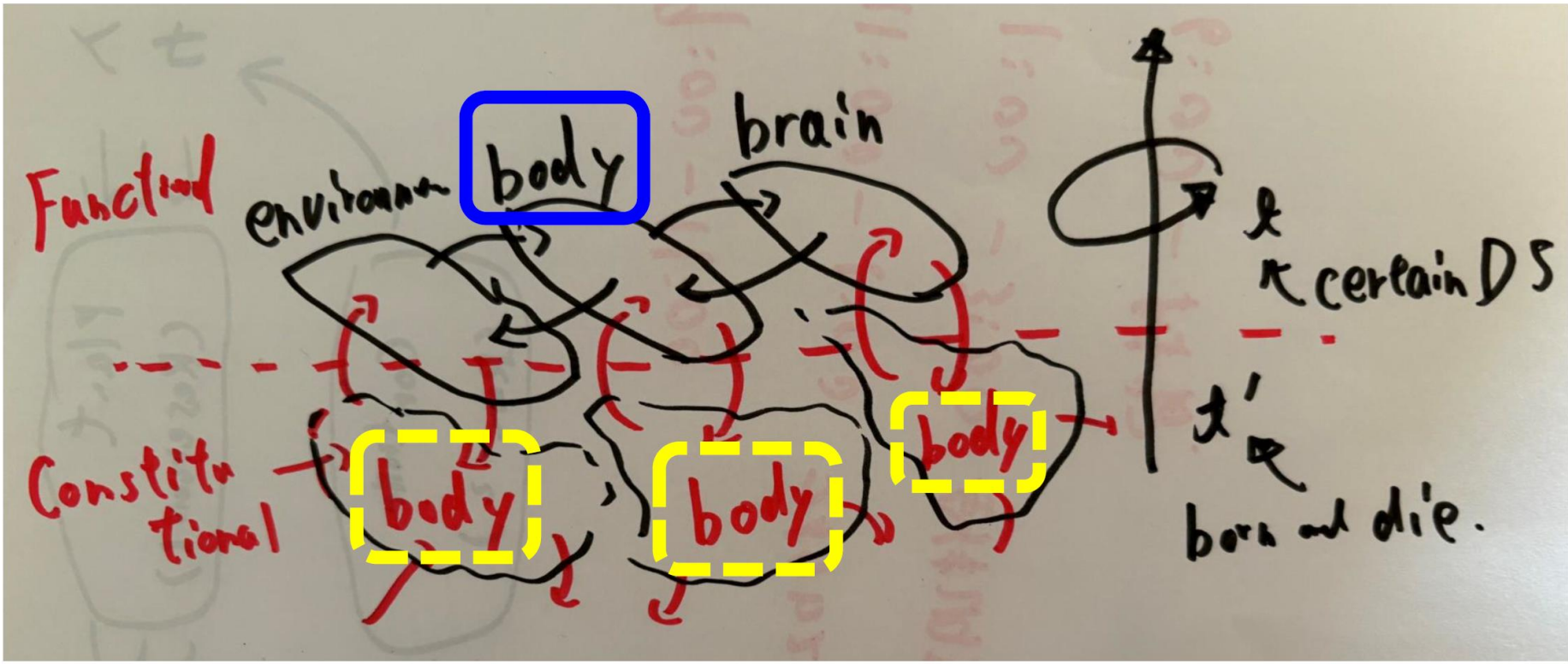


- **Constitutional layer**

physical material that constitutes a function, evolvable hardware, developmental aspect, edible and degradable, metabolism and energy

Interaction between **functional** and its **constitutional layers**!

Two bodies, two timescales



- No way to describe both simultaneously! (a single way including all is possible…but)
- System is latently exposed in different usage (a computer can be a desk, or even a frisbee!)

Mortal computation (Hinton 2022)

- Software and hardware is separated:
  Program won't die with hardware
  Immortal
  (make copies, reproduce)



Geoffrey Everest Hinton
2024Noble prize in physics

- Software and hardware is connected:
  Program dies with hardware
  Mortal
  (using intrinsic nonlinearity, save energy)

Physical reservoir is mortal!

Hinton, G. (2022). The forward-forward algorithm: Some preliminary investigations. *arXiv preprint arXiv:2212.13345*, *2*(3), 5.

Mortal computation (Hinton 2022)

"…More specifically the community has clung to the idea that the software should be separable from the hardware so that the same program or the same set of weights can be run on a different physical copy of the hardware. This makes the knowledge contained in the program or the weights **immortal**: The knowledge does not die when the hardware dies."

Separation of hardware/software = immortal!

Hinton, G. (2022). The forward-forward algorithm: Some preliminary investigations. *arXiv preprint arXiv:2212.13345*, *2*(3), 5.

Mortal computation (Hinton 2022)

The separation of software from hardware … makes it possible to study the properties of programs without worrying about electrical engineering…makes it possible to write a program once and copy it to millions of computers... If, however, we are willing to abandon immortality it should be possible to achieve huge savings in the energy required to perform a computation and in the cost of fabricating the hardware that executes the computation. We can allow large and unknown variations in the connectivity and non-linearities of different instances of hardware that are intended to perform the same task and then rely on a learning procedure to discover parameter values that make effective use of the unknown properties of each particular instance of the hardware. These parameter values are only useful for that specific hardware instance, so the computation they perform is **mortal**: it dies with the hardware.

Information processing is affected by physical substrates! = embodiment

Hinton, G. (2022). The forward-forward algorithm: Some preliminary investigations. *arXiv preprint arXiv:2212.13345*, *2*(3), 5.

Embodied Computation Manifesto

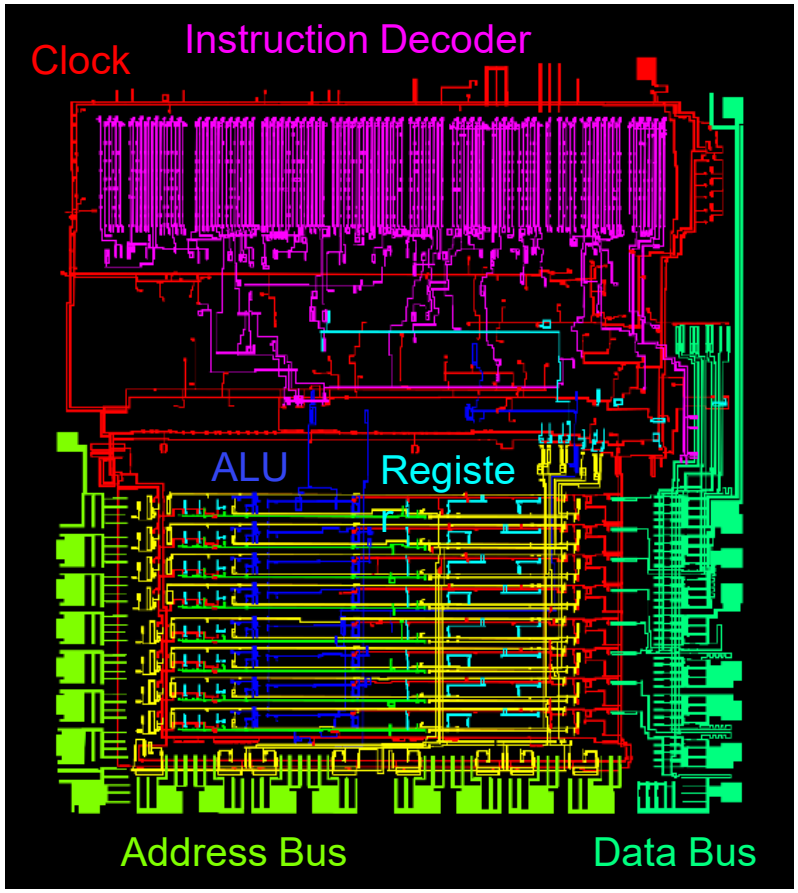
トヨタ・コンポン研　内臓付きAI研究チーム
（伊藤亜紗、一方井裕子、中嶋浩平、宮本道人、磯村朋子）



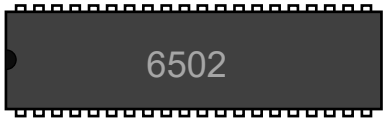
- Introduce “now” and “here” to the computers

TOYOTA KONPON RESEARCH INSTITUTE

Method: Visual 6502 Simulator



Simulator of MOS 6502 Processor



8-bit Processor

Launched in 1975 by MOS Technology

Used for computer (e.g., Apple I & II)

video games (e.g., NES & PC Engine)

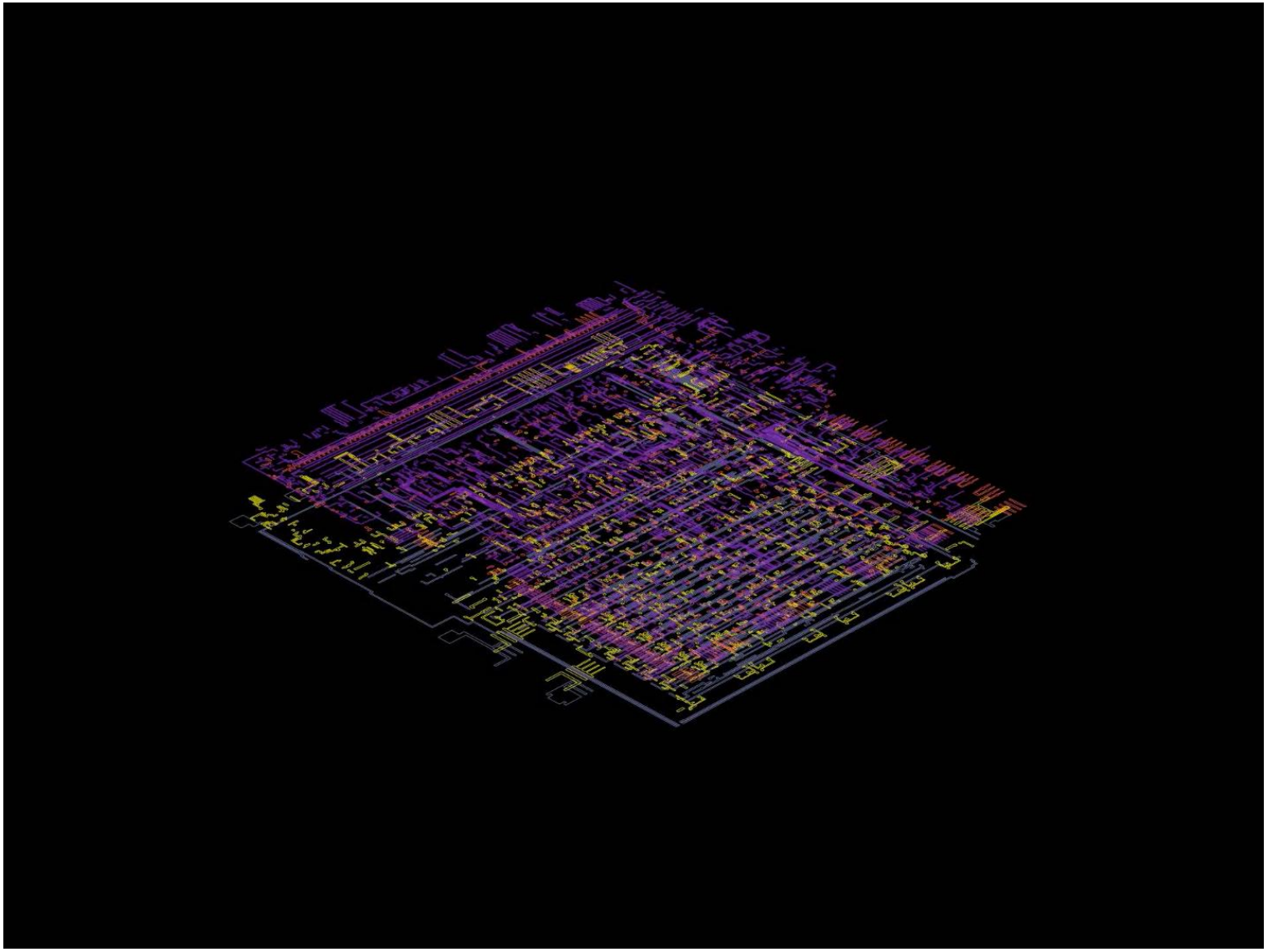
of signal lines: 1,704

of transistors : 3,510

Reproduce behavior of chip
on transistor level

CPU has complex dynamics = a physical body!

Mortal computation in CPU (in preparation)



CPU is more than “CPU”!

Family computer bugs



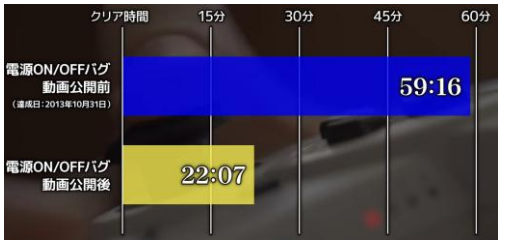
- Uses MOS6502!

Q. Huang, Y. Zhang, S. Ota, T. Kubota, K. Nakajima, Investigating Nintendo Family Computer Bugs: How Hardware-Level Perturbations Shape Software Behavior. In Artificial Life Conference Proceedings, pp. 975-983, 2026.

ドラクエ３のRTAにおける電源ON/OFFバグの発見







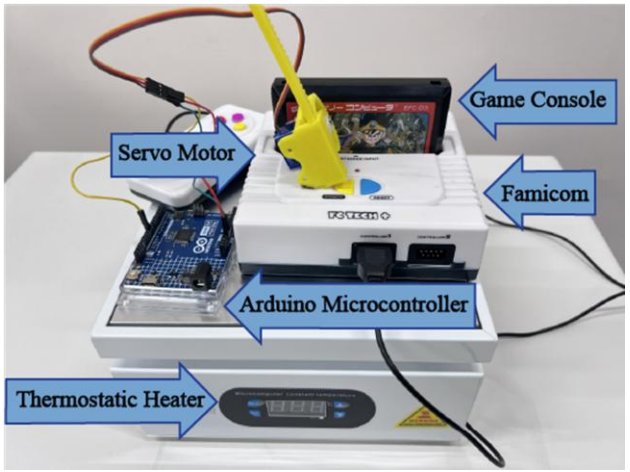
ドラクエIIIのRTA最速クリア記録は如何にして3分台になったのか…そこには電源バグが生み出した狂気の世界が存在した

https://www.youtube.com/watch?v=c3CdrA5vTNg

- ドラクエ３のReal time attack(RTA) の記録が発売から30年経って更新！
- セーブ中に電源をON/OFFしアイテムを入手する（レベルが最強に）
- 本体をホットプレートで加熱→発生確率上昇！

電源on/offバグの例





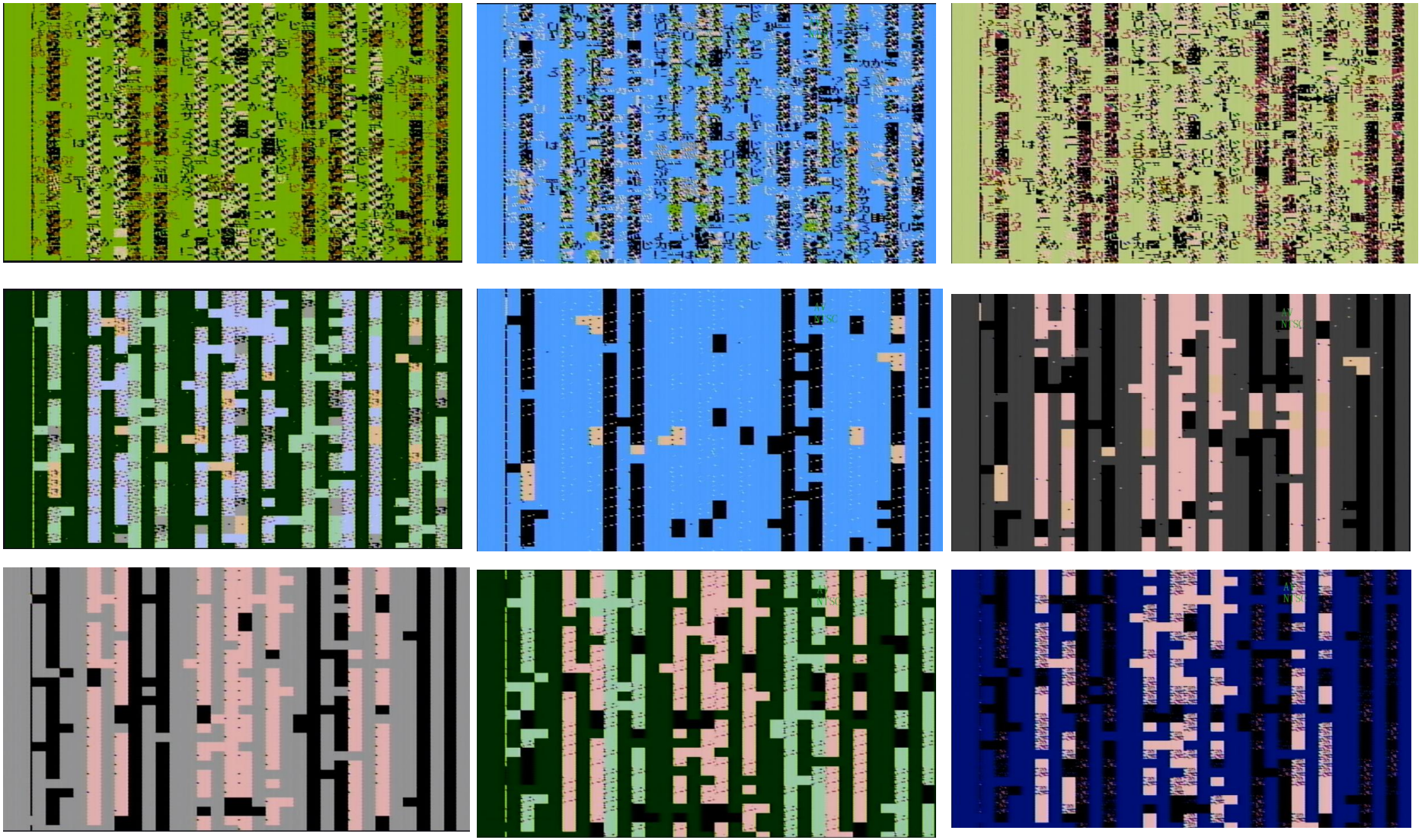
- 身体への定義外部の入力（摂動）が圧倒的なperformanceを生み出した例。
- 定義された入力の範囲では不可能！-> mortal computationに近しい

学生が新しいバグ発見!



glitch screens

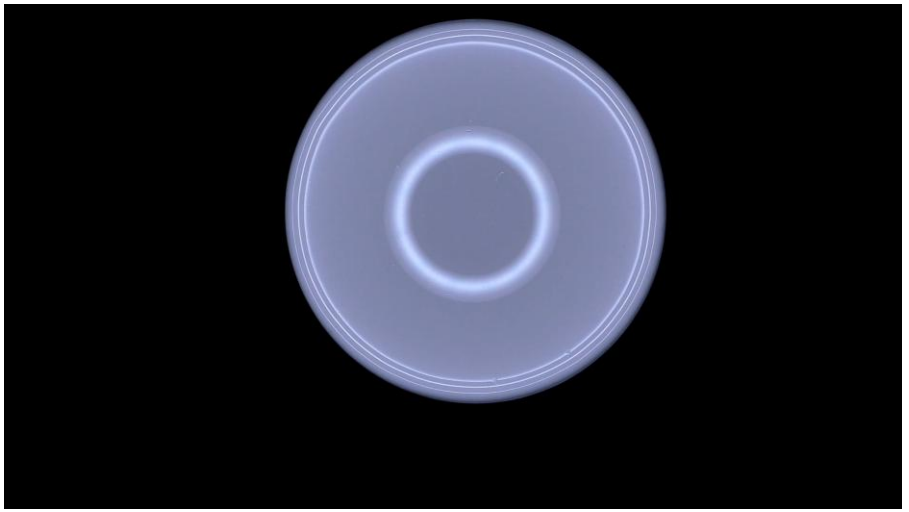
Q. Huang, Y. Zhang, S. Ota, T. Kubota, K. Nakajima, Investigating Nintendo Family Computer Bugs: How Hardware-Level Perturbations Shape Software Behavior. In Artificial Life Conference Proceedings 2026 (accepted).

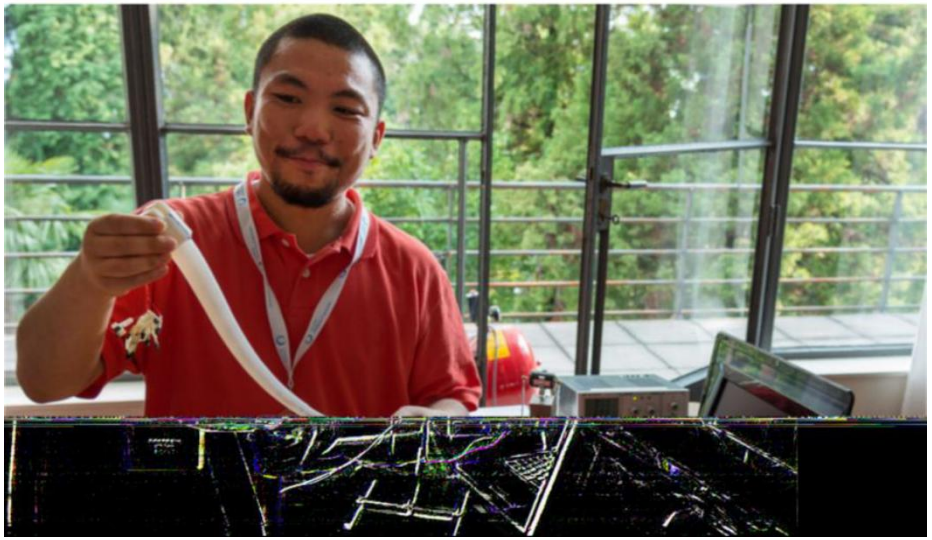


Measuring entropy of glitch screens!

In 2024, my PC broke down!

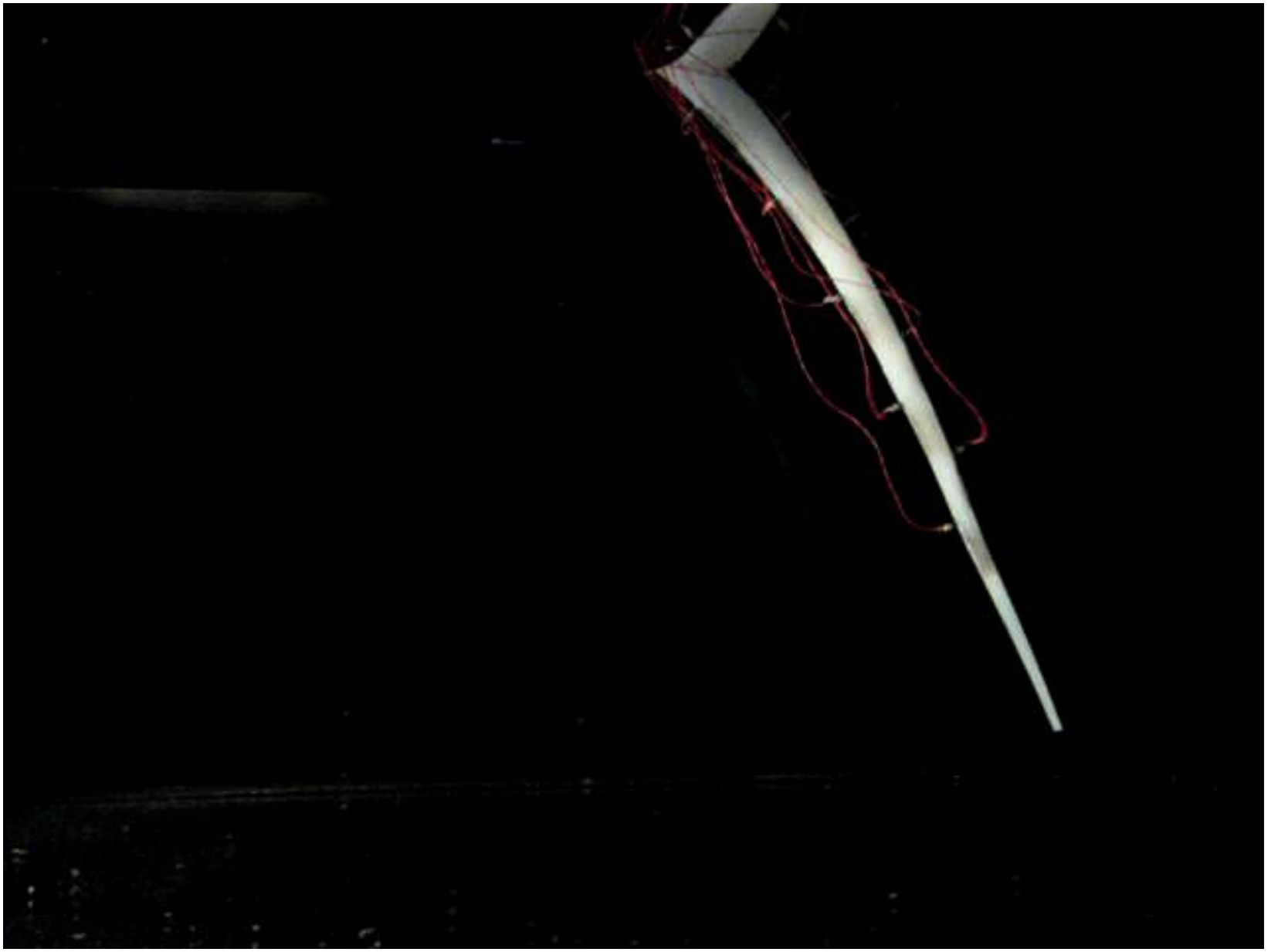




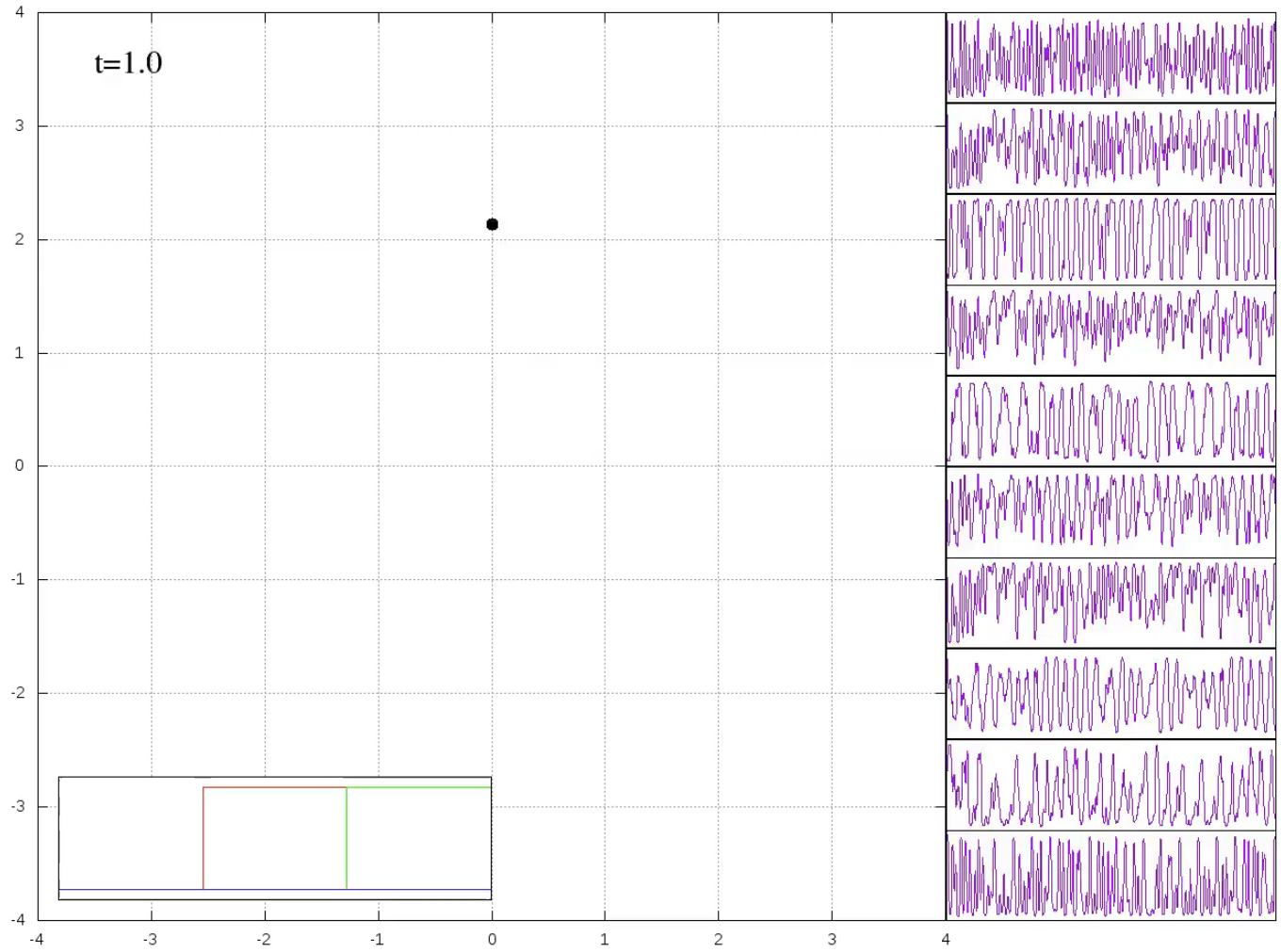


- Felt a bit of identity crisis!
- Memory was partially destroyed.
- Difficult to process memory with previous PC.

Physically perturbed recorded octopus arm

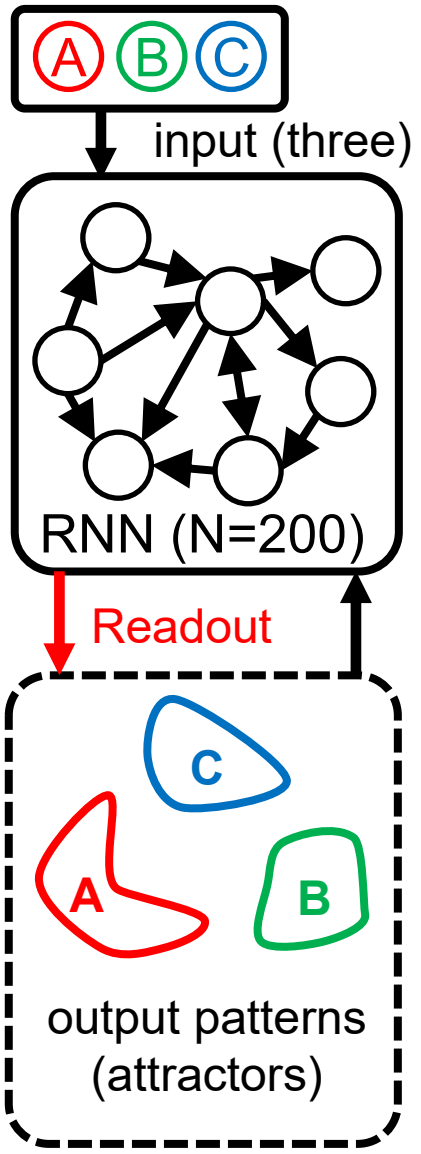


Physically perturbed recorded limit cycle embedding



Input commands (three)

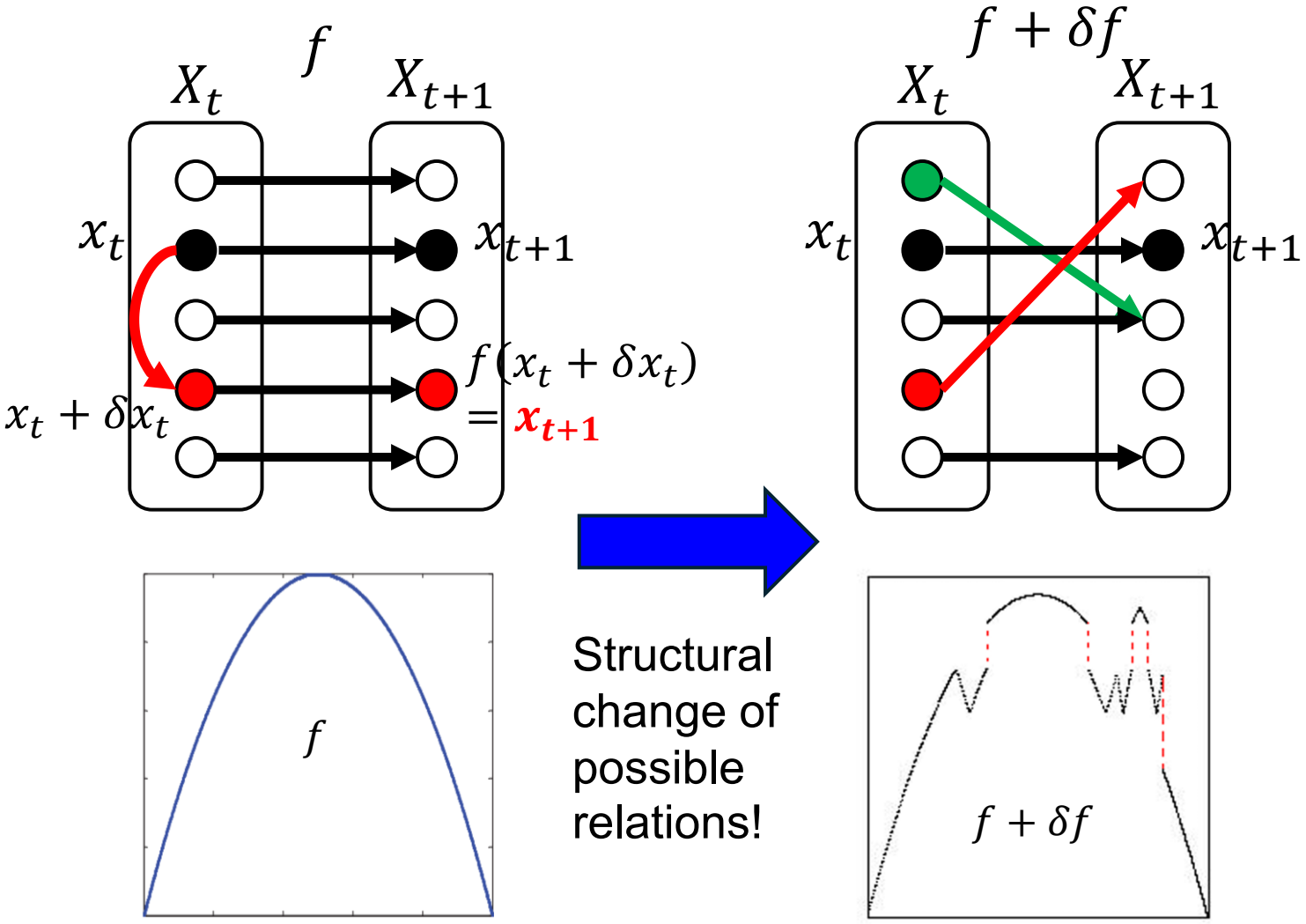
ESN dynamics
(from 10 nodes)



High Operability!

K. Inoue, K. Nakajima, Y. Kuniyoshi, Proceedings of NOLTA2018, pp. 412-414, 2018.

Consider $f + \delta f$, not $x + \delta x$!

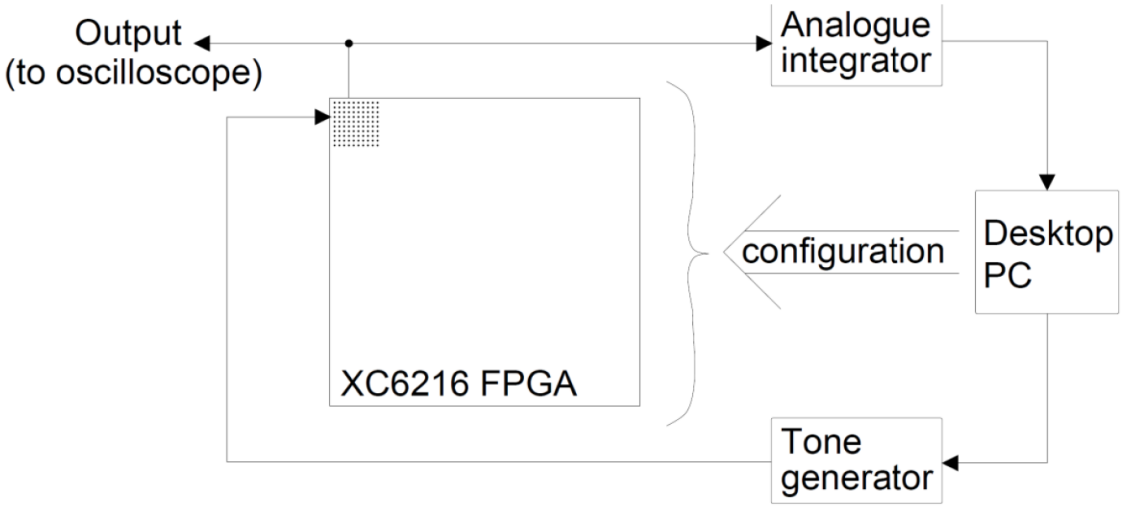


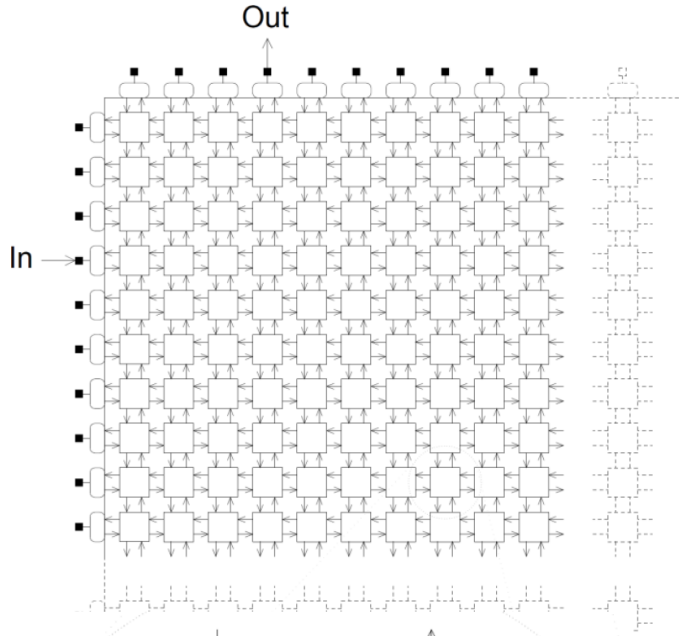
Gunji, Y. P., & Kamiura, M. (2004). Observational heterarchy enhancing active coupling. *Physica D: Nonlinear Phenomena*, *198*(1-2), 74-105.

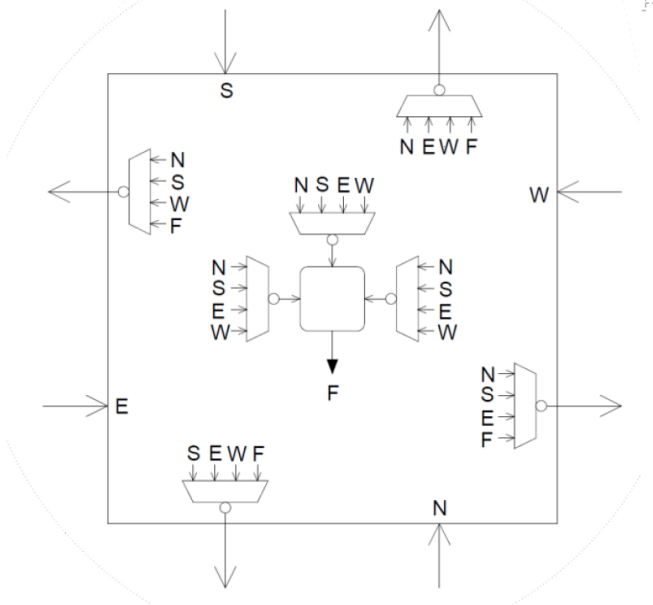
Perturbation to the constitutional layer affects functional layer!

Evolvable hardware 1

Thompson, A. (1997). An evolved circuit, intrinsic in silicon, entwined with physics. In *Evolvable Systems: From Biology to Hardware: First International Conference, ICES96 Tsukuba, Japan, October 7–8, 1996 Proceedings 1* (pp. 390-405). Springer Berlin Heidelberg.



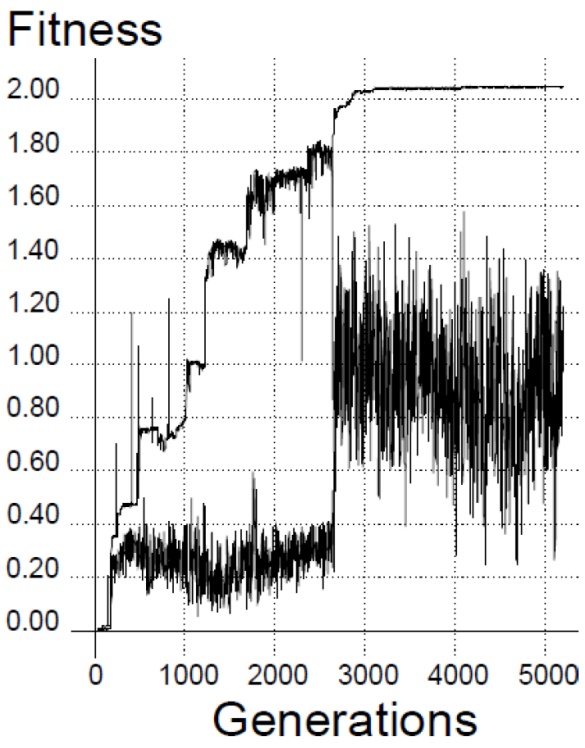


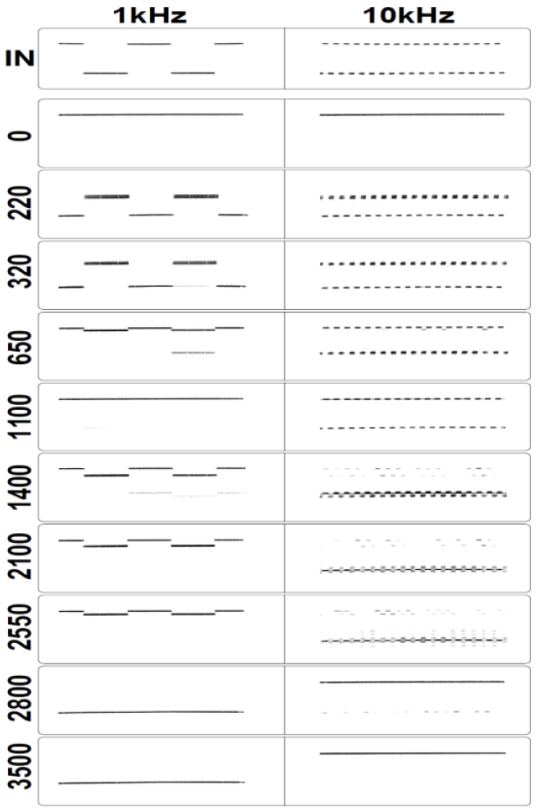


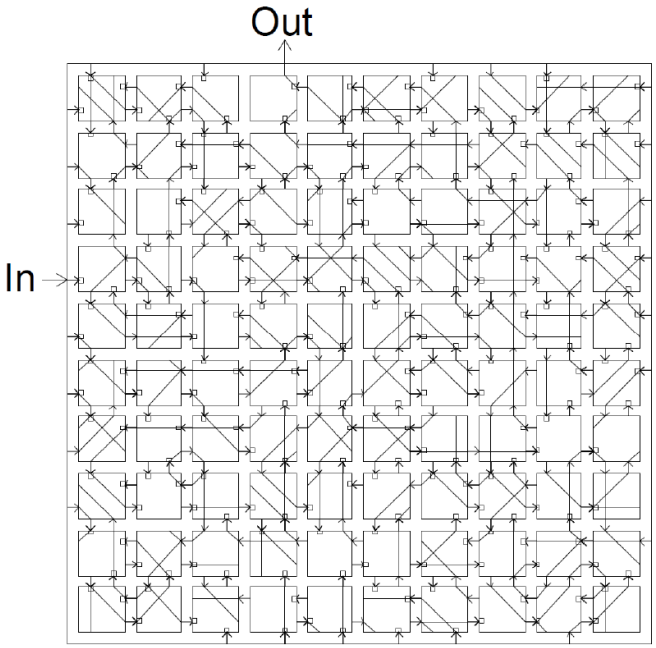
- If the square wave of 1kHz (10kHz) were presented, output should be 0 (1).
- Using GA to evolve connections and Boolean function for each cell.

Evolvable hardware 2

Thompson, A. (1997). An evolved circuit, intrinsic in silicon, entwined with physics. In *Evolvable Systems: From Biology to Hardware: First International Conference, ICES96 Tsukuba, Japan, October 7–8, 1996 Proceedings 1* (pp. 390-405). Springer Berlin Heidelberg.



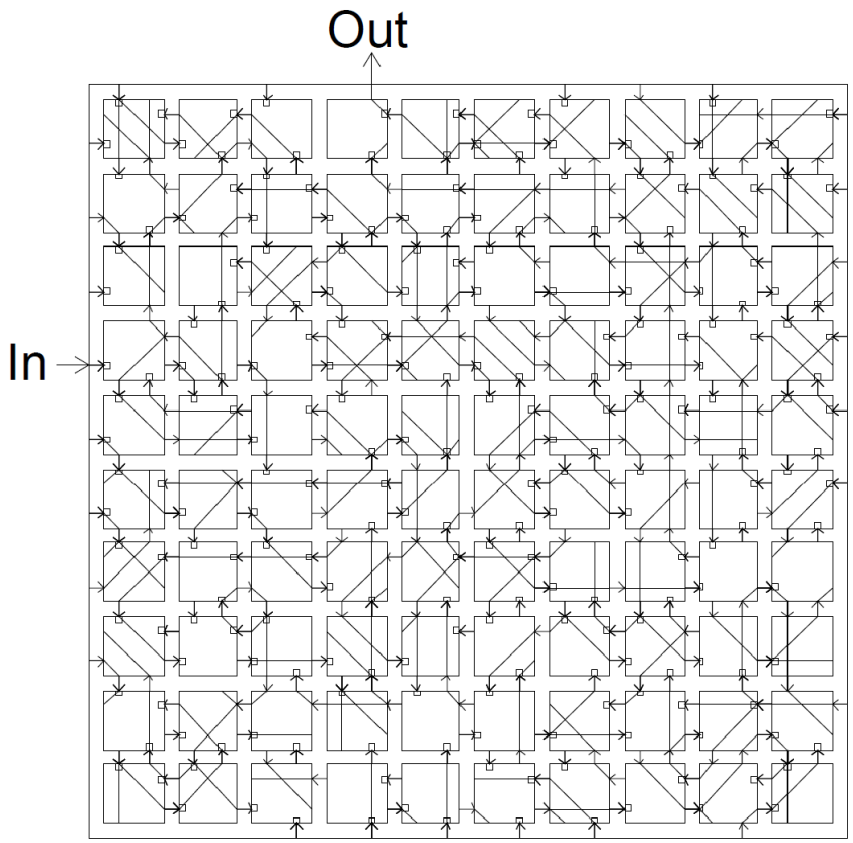


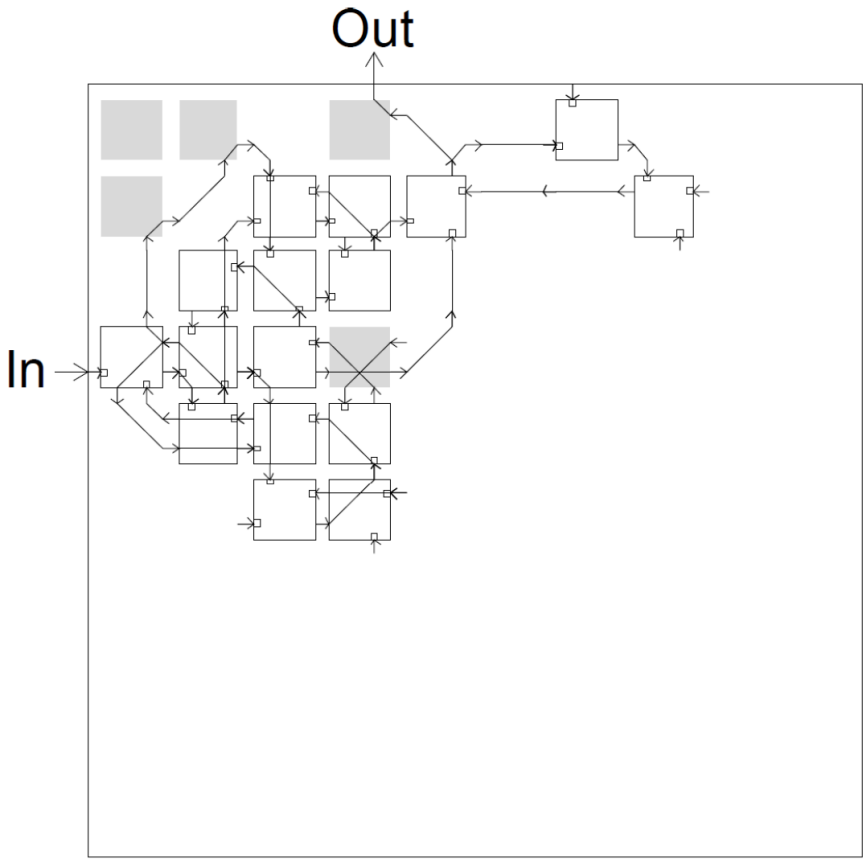


- 2-3 weeks of GA evolved successful agents.

Evolvable hardware 3

Thompson, A. (1997). An evolved circuit, intrinsic in silicon, entwined with physics. In *Evolvable Systems: From Biology to Hardware: First International Conference, ICES96 Tsukuba, Japan, October 7–8, 1996 Proceedings 1* (pp. 390-405). Springer Berlin Heidelberg.





- Undefined effect (interaction through the power-supply wiring, or electromagnetic coupling) has affected the functional layer due to physical body!

Edible robots and bio-degradable materials



Shintake, J., Sonar, H., Piskarev, E., Paik, J., & Floreano, D. (2017, September). Soft pneumatic gelatin actuator for edible robotics. In *2017 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)* (pp. 6221-6226). IEEE

T. Nagai, J. Shintake, Characterization of bio-degradable materials for soft robotics, in IEEE International Symposium on Micro-NanoMechatronics and Human Science (MHS), 2019.

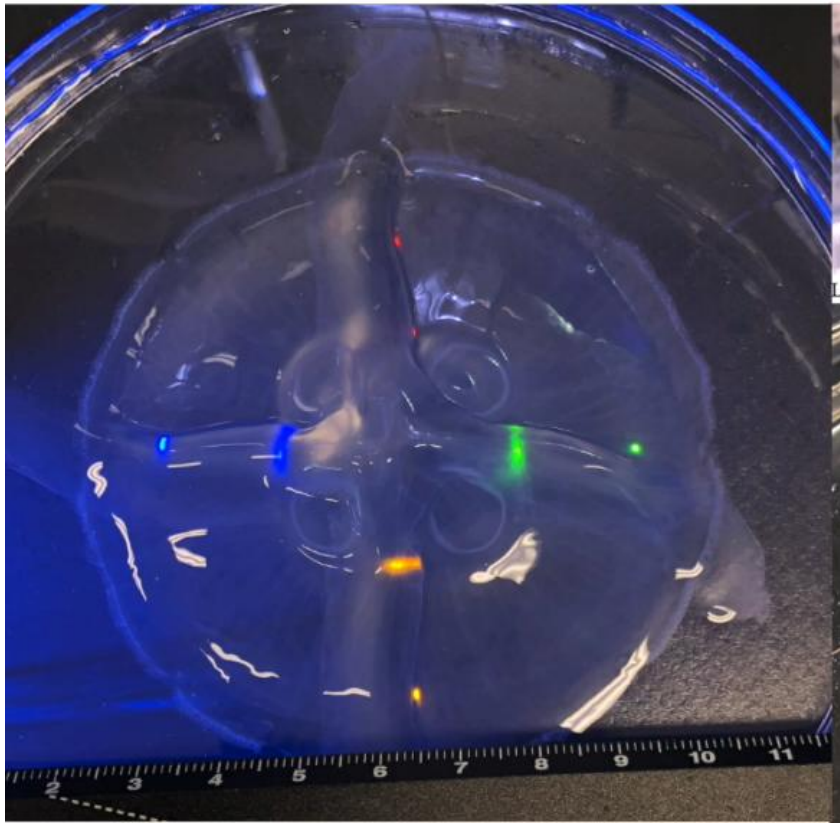


- Goes beyond sensorimotor coupling framework.
- Theoretical investigation is difficult!

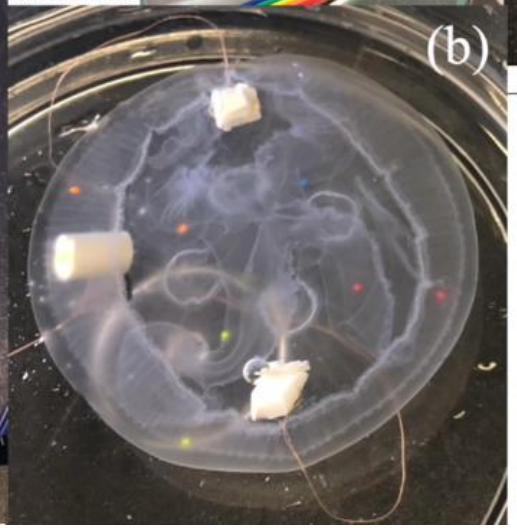
Jellyfish cyborg

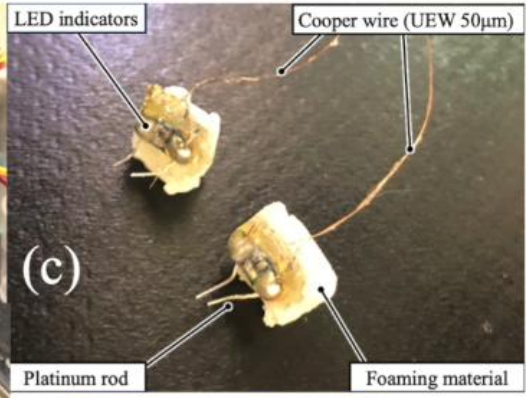
D. Owaki, M. Austin, S. Ikeda, K. Okuizumi, K. Nakajima, ***Nature Communications*** 16: 4642, 2025.

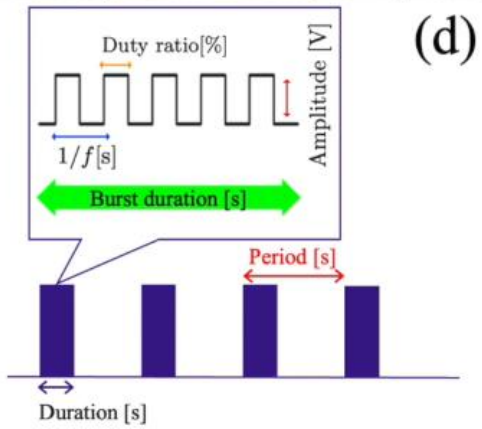
VIE Tags & Electrical Stimulation











https://www.youtube.com/watch?v=6kr4Il3FdMU

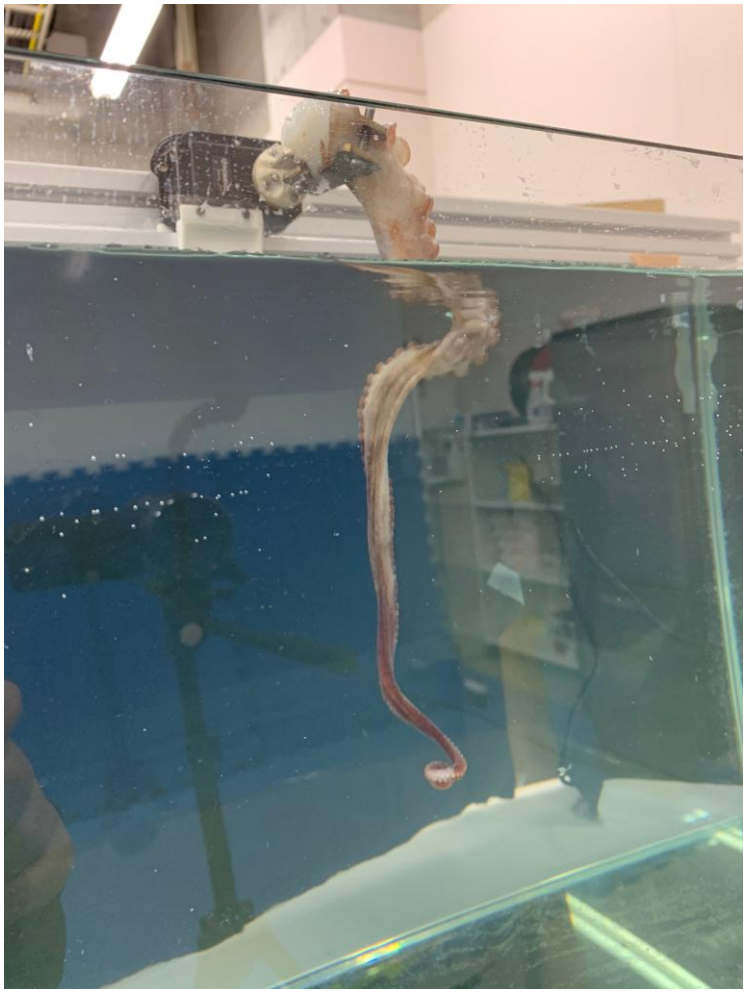
Controllability and spontaneity

Measuring transition from living to non-living…



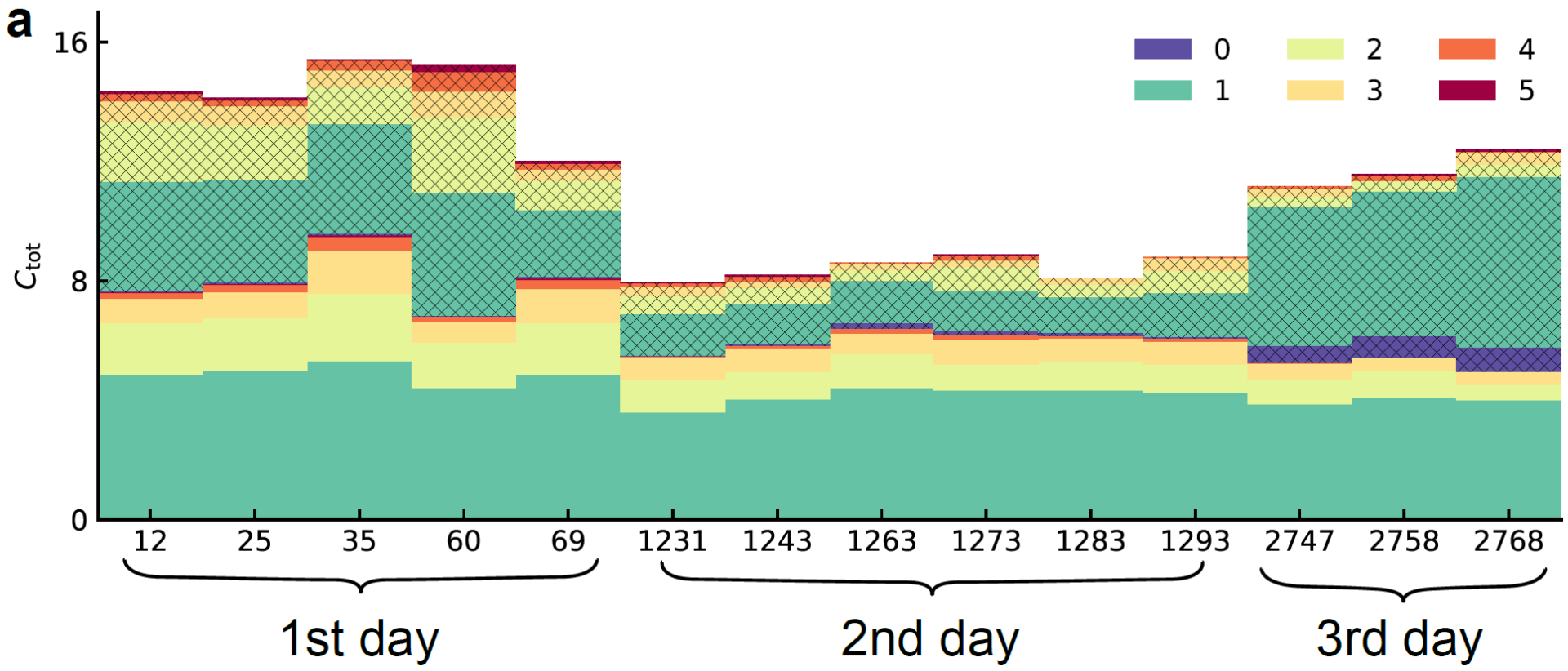






Characterizing its dynamics and computational anatomy

Transition of temporal IPC in dying process



- Just after cut off (self-oscillation)
- rigor mortis
- Materialization+degraded

B. Yu, T. Kubota, K. Nakajima, Computational anatomy of living and nonliving transitions: a case study on a real octopus arm. In ***Artificial Life Conference Proceedings*** 37, Vol. 2025, No. 1, p. 91, 2025.

What is embodiment?

It cannot simply be discarded once a robot has been built.＝
A science that examines the boundary conditions and constraints of systems

＝ Universality

- Something that inevitably accompanies the physical implementation of any abstract computation or function in this world.
  - ◆Physical reservoir computing: multiplexing and hacking computation
  - ◆Ensuring identity

$$f$$

- Something that induces change
  - ◆Changeability, adaptability

$$f + \Delta f$$

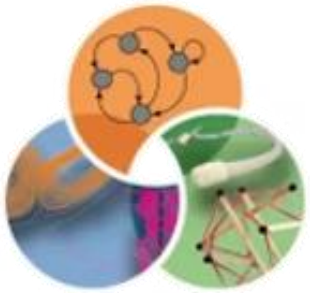
Form (use, redefinition, naming) by observer



Dynamic wholeness and intensity

reality

RC bootcamp released!



RC bootcamp

from Python/NumPy tutorial to cutting-edge reservoir computing study

eng: **https://rc-bootcamp.github.io**
jpn: **https://rc-bootcamp.github.io/ja**
github: **https://github.com/rc-bootcamp/rc_bootcamp**

Reference: K. Inoue, T. Kubota, Q. H. Tran, N. Akashi, R. Terajima, T. Kabayama, J. C. Guan, K. Nakajima, Reservoir computing bootcamp—from Python/NumPy tutorial for the complete beginners to cutting-edge research topics of reservoir computing, ***Chaos*** 36, 023109, 2026. (elected for a cover image)

Evaluation

- Reports (two or three)
- Attendance

Thanks!